



MAVIA BIN WASAD

01-134222-169

EEMAN TARIQ

01-134222-182

**A HYBRID MACHINE LEARNING FRAMEWORK FOR
CRYPTOCURRENCY PRICE PREDICTION**

Bachelor of Science in Computer Science

Supervisor: Dr. Sabina Akhtar

Department of Computer Science
Bahria University, Islamabad

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Certificate

We accept the work contained in the report titled "A HYBRID MACHINE LEARNING FRAMEWORK FOR CRYPTOCURRENCY PRICE PREDICTION", written by Mr. Mavia Bin Wasad and Ms. Eeman Tariq as a confirmation to the required standard for the partial fulfillment of the degree of Bachelor of Science in Computer Science.

Approved by:

Supervisor: Dr. Sabina Akhtar (Senior Assistant Professor)

Internal Examiner: _____

External Examiner: _____

Project Coordinator: Dr. Asim Ali (Fyp Coordinator CS Department)

Head of Department: Dr. Muhammad Khurram Ehsan (Head of Department Computer Science)

Abstract

This project presents a hybrid deep learning-based approach to price movement forecasting of cryptocurrencies based on a hybrid of Convolutional Neural Networks, Long Short-Term Memory networks, as well as BiLSTM models. The system is developed to be able to deal with the extremely volatile and nonlinear characteristics of cryptocurrency markets by learning both short term data and long-term dependencies in financial time series data. The combination of these two complementary structures will help the model identify local trends, relationships over time, and global contextual patterns in the data, leading to a more robust and reliable prediction framework.

Training is conducted on seven popular cryptocurrencies such as Bitcoin, Ethereum, Binance Coin, XRP, Cardano, Solana, and Dogecoin with a historical OHLCV data and sentiment data provided by the Fear and Greed Index. A detailed feature engineering method is used to derive significant indicators out of raw data, which allows the model to learn more about the market behavior. The system has good predictive performance especially in Bitcoin whereby high confidence directional accuracy is observed to be more than seventy percent, and this indicates the viability of the hybrid approach. Besides model development, the project also involves full-stack deployment with FastAPI backend and a React frontend, which lets users get real time predictions, visualize trends using interactive charts, and analyse model outputs in a user-friendly and intuitive environment.

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Quote

"Beliefs are hypotheses to be tested, not treasures to be guarded."

— Philip E. Tetlock

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Acronyms and Abbreviations

DeFi	Decentralized Finance
NFT	Non-Fungible Token
ML	Machine Learning
DL	Deep Learning
ANN	Artificial Neural Network
DNN	Deep Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
Bi-LSTM	Bidirectional Long Short-Term Memory
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from BiLSTMs
RoBERTa	Robustly Optimized BERT Approach
DistilBERT	Distilled BERT (a smaller, faster BERT model)
VADER	Valence Aware Dictionary and Sentiment Reasoner
FinBERT	Financial BERT (BERT fine-tuned for financial texts)
TF-IDF	Term Frequency Inverse Document Frequency
BoW	Bag of Words
NER	Named Entity Recognition
XAI	Explainable Artificial Intelligence
LIME	Local Interpretable Model-Agnostic Explanations
SHAP	SHapley Additive exPlanations
ARIMA	Autoregressive Integrated Moving Average

AR	Autoregressive
MA	Moving Average
SMA	Simple Moving Average
EMA	Exponential Moving Average
ES	Exponential Smoothing
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
SVR	Support Vector Regression
RF	Random Forest
GBM	Gradient Boosting Machine
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MPE	Mean Percentage Error
R^2	Coefficient of Determination
AUC	Area Under Curve
ROC	Receiver Operating Characteristic
F1	F1 Score
RSI	Relative Strength Index
MACD	Moving Average Convergence Divergence
ATR	Average True Range
VWAP	Volume Weighted Average Price
ROI	Return on Investment
OHLCV	Open High Low Close Volume
API	Application Programming Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
HTTPS	Hypertext Transfer Protocol Secure
SSL	Secure Sockets Layer
TLS	Transport Layer Security

CPU	Central Processing Unit
GPU	Graphics Processing Unit
UC	Use Case
GUI	Graphical User Interface
BTC	Bitcoin
ETH	Ethereum
XRP	Ripple

Chapter 1

Introduction

Over the past years, cryptocurrency has been one of the most disruptive financial innovations in the international financial market. Cryptocurrency markets are decentralized through blockchain technology, unlike a traditional financial system that is centralized and run by authorities. These markets are 24/7 with no specific time of trading, and they produce huge volume of real time information and appeal to investors, researchers and traders worldwide. The rapid expansion of cryptocurrencies like Bitcoin, Ethereum, Binance Coin, XRP, Cardano, Solana and Dogecoin has rendered them a significant field of research in the world of finance and technology. Yet, cryptocurrency markets are incredibly volatile and unpredictable, still, they are popular. The market forces have many factors that influence the fluctuation of prices such as demand and supply in the market, investor behavior, the economic condition of the world and technological advancements, as well as happenings outside the market. The nonlinear and dynamic behavior renders it very difficult to make accurate prediction.

Financial forecasting has traditionally been done by traditional statistical models. ARIMA and regression-based models are techniques which are based on assumptions of linearity and stability in data. Although effective in more structured and less volatile settings, these methods are ill adapted to the cryptocurrency markets where trends have complicated and ever-evolving patterns. This has created an increasing demand of more sophisticated methods that are capable of learning and adjusting to such environments. The development of artificial intelligence has led to the deep learning models that have demonstrated a great possibility of dealing with complex time series data. Convolutional Neural Networks are able to capture short term patterns and local features whereas the Long Short-Term Memory networks are structured in such a way that they are able to capture sequential dependencies across time. Moreover, BiLSTM models also include attention mechanisms, which allow determining global relationships within the data. All these models have their advantages, however, the use of one piece of architecture might restrict the ability to predict in general.

The concept of this project is to integrate various methods of deep learning into

a hybrid one. The system will combine CNN, LSTM and BiLSTM architectures by incorporating both short term variability and long term dependency of cryptocurrency data. This mixed method will increase the capacity of the model to recognize complex patterns and boost the performance of prediction.

Investor sentiment is another critical aspect that affects cryptocurrency markets. Fear and greed are some of the emotions that tend to affect market movements and can influence price movements greatly. In response, the project will use the Fear and Greed Index as an extra input, to apply sentiment information. This enables the model to take into account the numerical data as well as the market psychology resulting in more realistic and informed predictions. Along with predicting the price values, the system is oriented at determining the price movement direction, which is more feasible in the context of real world decision making. The project will be a complete system (data collection, preprocessing, feature engineering, model training, and deployment) based on the web-based interface. This guarantees the solution is not just sound theoretically but can be applied in practice as well.

1.1 Project Background

As the cryptocurrencies grow at a very fast pace, the issue of forecasting the price of digital assets has become a significant research topic. Cryptocurrency markets are extremely volatile and 24/7, and thus the changes in prices are hard to predict. Conventional forecasting techniques tend not to reflect the non-linear and complex nature of these markets. Cryptocurrencies have transformed into a worldwide financial phenomenon since the last decade since they were a niche idea. Decentralized digital currency started with the introduction of Bitcoin in 2009. Thousands of cryptocurrencies have been created since that time, each with its own features and applications. Cryptocurrencies have gained much popularity and as a result of this, a lot of research has been conducted in the area of price prediction and market analysis. Researchers seek to establish models on which they can comprehend and predict the complex behavior of markets whereas investors depend on correct predictions to make lucrative trading choices. Nevertheless, even with such attempts, the price of cryptocurrencies is a difficult task to predict because it is highly volatile, data are noisy, and external factors can affect the price. Conventional techniques can no longer be relied upon and new innovative techniques based on machine learning and deep learning are being considered to overcome these obstacles. [3].

1.2 Problem Description

Although the method of machine learning and deep learning has developed at an astonishing pace, a few practical issues regarding cryptocurrency price prediction continue to

be encountered. Cryptocurrency markets are highly dynamic and volatile, and therefore, it is hard to achieve consistent and reliable results using the existing systems. The existing methods have weaknesses in terms of their capability to capture the complexity of market behavior comprehensively. Current cryptocurrency prediction systems have several limitations, which diminish their effectiveness. Most models have used one algorithm, like Long Short-Term Memory (LSTM) networks or regression-based models, limiting their capability to encode various patterns found in time-series data. These models can therefore not be very effective when market conditions are very dynamic and fast changing.

The other key weakness is that most systems are interested in forecasting the specific prices values but not the price movement direction. In trading situations and in the real world it is more common that the important thing to predict is whether the price will rise or fall rather than the actual numerical value. Moreover, most existing models do not appropriately consider the sentiment data or ignore it. The cryptocurrency markets are highly affected by the investor psychology, and the failure to consider sentiment related information results in an incomplete analysis process and decrease in prediction accuracy.

Common Problems with Existing Cryptocurrency Price Prediction Approaches:

- **Limited Model Capability:** Most systems rely on a single model, which limits their ability to capture both short-term and long-term patterns in data.
- **Focus on Price Only:** Many models predict only price values and ignore the more practical task of predicting direction of movement.
- **Poor Sentiment Integration:** Existing approaches either ignore sentiment data or fail to incorporate it effectively into the prediction process.
- **Inability to Handle Market Complexity:** Traditional and single-model approaches struggle with the nonlinear and highly volatile nature of cryptocurrency markets.
- **Limited Practical Usability:** Lack of proper deployment and user-friendly interfaces makes these systems difficult to apply in real-world scenarios.

1.3 Project Objectives

The Hybrid Deep Learning Model to Cryptocurrency Price Prediction is intended to overcome the shortcomings of the current prediction systems by integrating various high-tech methods and enhancing usability.

The objectives are:

1. **Integrate Multiple Deep Learning Models:** Combine Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and BiLSTM architectures to capture both local patterns and long-term dependencies in cryptocurrency time-series data.
2. **Improve Prediction Accuracy:** Enhance the overall performance of the system by using a hybrid approach that leverages the strengths of different deep learning models.
3. **Incorporate Sentiment Analysis:** Integrate sentiment data using the Fear and Greed Index to include psychological and behavioral factors that influence market trends.
4. **Provide Dual Prediction Outputs:** Generate both price prediction and direction classification to support more practical and effective decision-making.
5. **Capture Complex Market Behavior:** Enable the system to handle the nonlinear and highly volatile nature of cryptocurrency markets through advanced learning techniques.
6. **Develop a User-Friendly Interface:** Design a web-based interface that allows users to easily view predictions, analyze trends, and interact with the system.
7. **Ensure End-to-End System Functionality:** Build a complete pipeline including data collection, preprocessing, model training, evaluation, and deployment for real-world usability.

1.4 Project Scope

The Hybrid Machine Learning Model for Cryptocurrency Price Prediction focuses on the following key components:

- **Data Collection and Preprocessing:** Gathering historical OHLCV data for seven major cryptocurrencies, including Bitcoin, Ethereum, Binance Coin, XRP, Cardano, Solana, and Dogecoin, and preparing it for model training and evaluation.
- **Feature Engineering and Data Integration:** Processing market data and incorporating sentiment information such as the Fear and Greed Index to enhance the quality of input features.

- **Hybrid Model Development:** Implementing a combination of Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and BiLSTM architectures to capture both short-term patterns and long-term dependencies in time-series data.
- **Daily Price Prediction:** Focusing on predicting daily cryptocurrency prices and direction of movement rather than high-frequency or intraday trading predictions.
- **Model Training and Evaluation:** Training the hybrid model using prepared datasets and evaluating performance using appropriate metrics to ensure reliability and accuracy.
- **Web-Based Application Development:** Designing and implementing a user-friendly web interface that allows users to view predictions, analyze trends, and interact with the system in real time.
- **System Deployment:** Deploying the complete system, including backend and frontend components, to provide accessible and practical prediction services.

The project combines these components for achieving an accurate and reliable prediction system for cryptocurrency markets. The system is designed for analytical purposes and not as a trading system for execution of trades or investment advice. Also, the modular design will support future upgrades to include additional data or better models.

Chapter 2

Literature Review

This chapter reviews what researchers and practitioners have already built in the space of cryptocurrency price prediction, with a particular focus on hybrid deep learning models that combine convolutional feature extraction, sequential time-series modeling, and attention-based mechanisms. Rather than simply cataloguing existing work, the aim here is to build a solid theoretical foundation for the proposed Hybrid CNN-LSTM + BiLSTM model, while also tracing how forecasting approaches have evolved and what challenges remain in predicting markets as volatile as cryptocurrency.

Early work in this area leaned heavily on classical statistical tools like ARIMA, exponential smoothing, and GARCH [1, 4]. These methods brought a useful mathematical framework to time-series forecasting, but they came with a critical assumption baked in: that the data is roughly linear and stationary. Cryptocurrency markets, as anyone who has watched a price chart for more than five minutes will know, are neither. This fundamental mismatch meant these models often struggled to produce forecasts that held up in practice.

As machine learning and deep learning matured [5, 6], the field shifted toward more flexible approaches. Regression algorithms, random forests, and especially LSTM networks became widely adopted for financial forecasting. LSTMs brought a real improvement in handling temporal dependencies, but even they had blind spots: capturing both fine-grained local patterns and broader long-range relationships at the same time proved difficult. Hybrid models combining CNN and LSTM were introduced to bridge this gap, allowing networks to extract local trends and learn sequential structure simultaneously. More recently, BiLSTM-based architectures have attracted attention for their ability to model long-range dependencies through self-attention.

Another thread running through recent literature is the role of sentiment. Cryptocurrency prices do not move in a vacuum, and market psychology has a measurable impact on price behavior. Indicators like the Fear and Greed Index have been used to bring that psychological dimension into predictive models. However, studies also flag a recurring pitfall: poor feature engineering and redundant inputs can hurt more than they help.

On balance, recent research points toward hybrid CNN-LSTM + BiLSTM architectures paired with sentiment-aware features as the most promising direction, though challenges around multi-coin generalization, evaluation consistency, and deployment in real-world systems remain largely open.

2.1 Cryptocurrency Prediction and Its Applications

Cryptocurrency markets have expanded at a remarkable pace since Bitcoin’s launch in 2008, growing into a complex, highly active financial ecosystem that operates around the clock with no centralized authority. This 24/7 nature, combined with dramatic price swings and the heavy influence of market sentiment, makes cryptocurrency prediction one of the harder forecasting problems in finance today.

The earliest attempts to tackle this problem borrowed directly from traditional finance, reaching for tools like ARIMA, exponential smoothing, and GARCH-based models. These approaches assume that price data follows linear, stable patterns over time, which is a reasonable assumption in many financial markets but a poor fit for cryptocurrencies. The decentralized, largely unregulated nature of these markets amplifies uncertainty and leaves prices highly sensitive to external shocks, speculative surges, and news events that no statistical model built on historical averages can easily anticipate.

As the field matured, researchers began turning to machine learning and deep learning in search of tools capable of modeling the non-linear patterns and temporal dependencies that statistical methods missed. Regression-based approaches, random forests, and LSTM networks each brought genuine improvements in forecasting accuracy, but single-model pipelines still fell short when asked to simultaneously capture short-term market noise and longer-range structural trends in volatile conditions.

The most recent wave of research has moved toward hybrid and sentiment-aware architectures. Combining CNN, LSTM, and BiLSTM components into unified models has produced strong results by integrating feature extraction, sequential learning, and attention in a single framework [8]. Sentiment signals such as the Fear and Greed Index and social media activity have also become standard additions to the feature set. Even so, challenges around generalization across assets, avoiding feature redundancy, and deploying these systems reliably in production settings continue to limit how far the field has progressed.

2.1.1 Comparison Between Traditional Models and Deep Learning

Traditional forecasting techniques tend to hit a ceiling when asked to capture the kind of complex, nonlinear dynamics that define financial time-series data. In markets as volatile

as cryptocurrency, this ceiling is reached quickly, and models built on assumptions of linearity or stationarity often fail to keep pace with how rapidly conditions change.

Deep learning approaches do not carry the same restrictive assumptions. They learn directly from data, which makes them naturally better suited to dynamic financial environments where patterns shift and relationships evolve over time. They can model both sequential dependencies across time and spatial or local features within a time window, something traditional models simply cannot do within a single framework.

Key deep learning methods used in cryptocurrency prediction include:

- **Recurrent Neural Networks (RNNs):** Designed to process sequential data by maintaining memory of previous inputs, making them a natural starting point for time-series forecasting.
- **Long Short-Term Memory (LSTM) Networks:** An improved variant of RNNs that addresses the vanishing gradient problem and can hold onto relevant information across much longer time horizons, making them effective for modeling extended price trends.
- **Gated Recurrent Units (GRU):** A leaner alternative to LSTM that reduces computational overhead while retaining strong sequence learning capabilities.
- **Convolutional Neural Networks (CNN):** Originally developed for image recognition, CNNs adapt well to financial time-series by extracting local patterns and short-term trends from raw price data, which improves feature representation before passing data to sequential models.
- **Hybrid Models (CNN-LSTM):** Combine the local feature extraction strength of CNNs with the sequential learning capability of LSTMs, resulting in more comprehensive forecasting architectures that outperform either model used alone.

The tradeoff with deep learning is real: these models are computationally heavier and require larger datasets to train reliably. But when those conditions are met, they consistently outperform traditional statistical approaches in cryptocurrency price prediction, largely because they can learn complex, nonlinear relationships rather than assuming the data fits a predefined structure [4].

2.1.2 Challenges in Cryptocurrency Prediction

Cryptocurrency price prediction is genuinely hard, and not just because markets move fast. The core difficulty is that prices are shaped by an unpredictable mix of sentiment, breaking news, speculative trading behavior, and macroeconomic events, all of which introduce substantial noise into the data. When prices can swing dramatically in response

to a single tweet or regulatory announcement, even well-designed models trained on clean historical data will struggle to keep up with real market dynamics.

Traditional statistical methods tend to break down in these conditions. They are built for environments where trends follow relatively predictable patterns, and cryptocurrency markets rarely cooperate. Machine learning models handle this better, using historical prices, technical indicators, and sentiment-derived features to build a richer understanding of market behavior. Among these, ensemble approaches have shown particular promise.

Qureshi et al. [5] evaluated models like Random Forest, XGBoost, and Gradient Boosting specifically for cryptocurrency forecasting and found that combining multiple learners consistently outperformed any single algorithm, especially during periods of high volatility. Their results also highlighted the practical importance of hyperparameter tuning: well-tuned ensemble models were not only more accurate but also more robust, reducing overfitting without sacrificing adaptability. The overall takeaway from that line of work is clear: machine learning approaches offer meaningfully better adaptability and predictive power than traditional methods when markets are highly dynamic, and ensemble strategies amplify those advantages.

2.1.3 Deep Learning Approaches

Deep learning has established itself as the most capable class of methods for cryptocurrency price prediction, largely because of its ability to uncover complex temporal and spatial patterns in markets that move unpredictably. Different architectures contribute in complementary ways. LSTM networks excel at capturing long-term sequential dependencies, allowing the model to remember how price conditions unfolded over time. CNNs bring a different strength: they are particularly good at picking up short-term local patterns and fine-grained trends within a time window. BiLSTM models add a layer of global awareness through self-attention, enabling the network to identify important relationships between time steps that may be far apart in the sequence. Used together, these architectures cover a broader range of market dynamics than any single model can.

Moutsinga and Pindza [6] put this directly to the test, studying LSTM, GRU, and Bidirectional LSTM networks across Bitcoin, Ethereum, and Litecoin price prediction tasks. BiLSTM consistently came out on top, producing lower error rates on metrics like RMSE and MAPE compared to both standard LSTM and GRU. The study also made a point worth noting for anyone building these systems: careful hyperparameter tuning and rigorous data preparation made a meaningful difference in forecast quality, not just marginal tweaks at the edges. These findings reinforce the case for bidirectional architectures in cryptocurrency forecasting, particularly in markets where the relationship between past and future price behavior is anything but straightforward.

2.1.4 Sentiment Analysis in Cryptocurrency Markets

One of the more interesting developments in cryptocurrency prediction research has been the growing recognition that market psychology matters as much as price history. Emotions, news cycles, and speculative behavior all leave measurable traces in how prices move, and the Fear and Greed Index is one of the more practical tools for quantifying that broader sentiment. By incorporating such signals, models gain context that raw OHLCV data simply cannot provide on its own.

Patel et al. [7] built on this premise by developing a deep learning framework that combined Bi-LSTM modeling with real-time Twitter sentiment scores to forecast hourly Bitcoin prices. The setup was straightforward: collect sentiment data from Twitter alongside price history, feed both into a bidirectional recurrent network, and see what happens. The results were clear — including sentiment significantly improved forecasting accuracy over models trained purely on historical price data.

What this work demonstrates is that social media is not just background noise. When modeled properly, it functions as a leading signal that can meaningfully sharpen predictions in volatile markets like cryptocurrency, where crowd psychology often drives price action more than fundamentals do.

2.1.5 Prediction and Market Behavior

There is an interesting feedback loop at work in financial forecasting that is easy to overlook. As predictive models become more sophisticated and widely adopted, the traders and investors using them begin to adjust their behavior accordingly, which in turn reshapes the very market dynamics those models were trained to capture. This means that cryptocurrency price prediction is not a problem you solve once and move on from; it is a moving target that evolves as market participants adapt.

The practical implication is straightforward: models need to be retrained regularly to stay relevant. A model that performed well six months ago on data from a specific market regime may struggle when conditions shift, not because the architecture is flawed, but because the patterns it learned no longer reflect how the market is behaving.

Zou and Herremans [8] offer a compelling example of how to work with this reality rather than against it. Their PreBit model integrates Twitter-based FinBERT embeddings with technical indicators and broader market data to predict extreme price movements in Bitcoin. By drawing on a domain-specific BiLSTM pretrained on financial text, they were able to significantly improve accuracy in detecting sudden, large price swings compared to models relying on historical price data alone. The finding underscores a broader point: architectures that can incorporate external context, particularly sentiment signals from financial text, bring meaningful predictive power to markets where sharp moves are often driven by information that never shows up in a price chart.

2.1.6 Data, Feature, Model Optimization, and Deployment Challenges

Building an effective cryptocurrency prediction system involves navigating a set of interconnected challenges that stretch well beyond choosing the right model architecture. The underlying data itself is often the first obstacle. Financial datasets tend to be noisy and incomplete, with missing values, extreme price spikes, and inconsistencies that can quietly undermine model performance if not handled carefully. Good feature engineering is essential here, not as an optional enhancement but as a prerequisite for meaningful learning.

Beyond data quality, the choice of model architecture and the process of tuning its hyperparameters have an outsized effect on outcomes. Different model families perform differently depending on the asset and the market regime, which means there is no universally optimal configuration. On top of that, responsible deployment requires more than just a well-trained model. Results need to be reproducible, validation has to be rigorous, and the system must be monitored continuously to catch performance degradation before it becomes a problem.

Interpretability is another dimension that matters considerably in high-stakes financial contexts. Goodell, Jabeur, Saadaoui, and Nasir [9] addressed this directly by proposing an explainable AI framework for Bitcoin price prediction that combines SHAP (SHapley Additive exPlanations) with game-theoretic feature selection. Their model does not just forecast price movements; it surfaces which input variables, macroeconomic indicators, past returns, and others, are driving those predictions and to what degree. The SHAP analysis revealed specific inflection points where certain features exerted disproportionate influence on the model’s behavior. For investors and analysts working in volatile markets, this kind of transparency is not just academically interesting; it is what makes a forecasting system genuinely trustworthy and usable.

2.2 Comparative Study

This section presents a structured review and comparison of existing research, tools, and frameworks related to cryptocurrency price prediction and hybrid machine learning models. The broader literature consistently points in the same direction: hybrid models that combine CNN, LSTM, and BiLSTM architectures outperform standalone approaches [6, 7]. These hybrid designs bring together local feature extraction, sequential dependency modeling, and global attention mechanisms, giving them a more complete view of market dynamics than any single model can achieve. As a result, they tend to generalize better across multiple cryptocurrencies and hold up more reliably under volatile conditions. The proposed system in this project is built directly on these findings, extending the

hybrid framework with the goal of improving both accuracy and robustness in real-world forecasting scenarios.

What the comparative analysis makes clear is that most existing studies either rely on a single model or work with a narrow feature set, rarely combining both. There is an evident gap in the literature when it comes to unified hybrid systems that bring CNN, LSTM, and BiLSTM architectures together with multi-source, sentiment-aware data in a single deployable framework [10, 12]. The proposed system is designed specifically to fill that gap, improving not just accuracy but also robustness and real-world usability.

Table 2.1: Comparative Study of Existing Work

Ref	Title	Key Insight	Project-Relevant Gap
1	Bitcoin Volatility Analysis [1]	GARCH models effectively estimate Bitcoin volatility for risk assessment	Limited to Bitcoin and statistical methods; no hybrid ML-DL approach
2	Bitcoin & Ethereum Bubble Analysis [2]	Identifies historical bubble periods in BTC and ETH markets	Focused on past events; lacks predictive modeling capability
3	Crypto Prediction via Sentiment Analysis [3]	Sentiment improves short-term prediction accuracy	Uses only sentiment; no hybrid or multi-source integration
4	Financial Time Series Prediction [4]	Compares classical statistical and ML methods	Not crypto-specific; lacks deep learning integration
5	ML Models for Crypto Forecasting [5]	Ensemble ML models like XGBoost improve accuracy	No deep learning or hybrid architecture
6	Deep Learning for Crypto Forecasting [6]	LSTM and GRU capture temporal dependencies effectively	Lacks hybrid integration with other models
7	Bi-LSTM & Twitter Sentiment Prediction [7]	Sentiment + Bi-LSTM improves hourly Bitcoin forecasting	Single-crypto focus; no multi-asset generalization
8	Multimodal Crypto Prediction [8]	NLP embeddings improve extreme movement detection	Focused on extreme events only; incomplete pipeline
9	Explainable AI for Bitcoin Forecasting [9]	Improves interpretability of predictions	Limited to Bitcoin; lacks hybrid model structure
10	Altcoin Forecasting Using LSTM [10]	LSTM captures trends in altcoin price data	No hybrid ML-DL or sentiment integration
11	Ensemble Methods for Crypto Prediction [11]	Ensemble models reduce error and improve stability	No deep learning integration
12	Real-Time Cryptocurrency Forecasting [12]	Enables dynamic stream-based prediction	Limited feature integration and modeling depth
13	Technical Indicator Analysis [21]	RSI, MACD, moving averages improve short-term trends	Only technical indicators; no hybrid approach
14	Feature Engineering for Crypto [22]	Market + sentiment + volume improves performance	No real-time or hybrid architecture
15	Extreme Price Movement Detection [8, 12]	Detects sudden spikes using ML + NLP models	Focused only on extreme events; not general forecasting

Chapter 3

Requirement Specifications

Any predictive system operating in a volatile, high-stakes environment like cryptocurrency markets can only be as good as the requirements it is built on. Vague or incomplete requirements do not just slow down development; they show up later as inaccurate predictions, brittle data pipelines, security gaps, or a system that simply cannot deliver insights when users need them most. All of these failures have direct consequences for financial decision-making.

For this project, defining requirements was treated as foundational work rather than a procedural formality. The goal of the Hybrid ML Cryptocurrency Price Prediction System is to combine machine learning, deep learning, and sentiment analysis into a unified architecture capable of delivering meaningful price predictions across multiple cryptocurrencies. Achieving that requires requirements that address both the core functional pieces, such as data collection, preprocessing, model training, feature engineering, and real-time prediction, and the non-functional dimensions that determine how well those pieces actually work in practice, including performance, security, usability, and reliability under rapidly shifting market conditions.

A well-structured requirements specification for this kind of system serves several purposes worth making explicit:

- It creates a shared understanding among developers, users, and stakeholders about what the system is supposed to do, what it will not do, and what constraints it operates under.
- It provides a coherent roadmap for the design and implementation phases, so that data pipelines, model components, and prediction modules can be developed and connected without constant renegotiation.
- It establishes concrete, measurable criteria for evaluating prediction accuracy, model performance, latency, and system response time, giving the team something to test against rather than a vague sense of “good enough.”

- It keeps scope drift in check by clearly defining what the system is expected to do and where its boundaries lie.

Gathering these requirements involved a close look at the limitations that show up repeatedly in existing forecasting approaches: over-reliance on a single model, failure to incorporate sentiment signals, and insufficient handling of the extreme volatility that characterizes cryptocurrency markets. It also meant thinking carefully about who actually uses these systems, traders, analysts, researchers, and investors who need fast, accurate, and interpretable outputs without having to understand the internals of a deep learning pipeline.

The requirements are organized into two categories. Functional requirements describe what the system must do: collect market data, perform time-series forecasting, analyze sentiment, train hybrid models, and produce real-time price predictions. Non-functional requirements describe how well the system must do these things, covering reliability, performance, security, usability, and scalability. Together, they form the specification that guides everything that follows.

3.1 Existing System

3.1.1 Cryptocurrency Price Prediction Models

Traditional cryptocurrency price prediction systems have largely relied on statistical tools like ARIMA, GARCH, and linear regression to analyze historical time-series data. These approaches have a certain appeal: they are well-understood, computationally lightweight, and relatively easy to interpret. Under controlled conditions with structured, stable data, they can produce useful short-term forecasts.

The problem is that cryptocurrency markets are neither controlled nor stable. Price behavior in these markets is highly volatile and driven by non-linear dynamics that statistical models built on assumptions of linearity and stationarity simply cannot capture. When conditions change sharply, as they often do, these models tend to lag behind reality rather than anticipate it.

Strengths

- Makes good use of historical price data to identify basic trends and temporal patterns.
- Simpler and computationally lighter than deep learning alternatives.
- Produces interpretable outputs that are straightforward to analyze and explain.
- Provides a useful baseline for benchmarking more advanced machine learning and deep learning approaches.

Limitations

- Cannot capture the non-linear and highly complex relationships present in cryptocurrency data.
- Performance degrades significantly in volatile, fast-moving market conditions.
- Ignores external factors like market sentiment, social media influence, and macroeconomic news.
- Relies on data stationarity assumptions that rarely hold in real-world cryptocurrency markets.

3.1.2 Time-Series Forecasting Approaches

Time-series forecasting approaches in cryptocurrency prediction focus on analyzing sequential historical price data to project future market behavior. The goal is to capture temporal dependencies, trends, and short-term fluctuations that characterize volatile financial markets. Modern implementations typically reach for LSTM and BiLSTM networks because of their ability to learn long-term dependencies across sequences, something simpler statistical models cannot do. That said, even these models have limits. Used in isolation, they struggle to simultaneously represent both fine-grained local patterns and the broader global relationships that matter in highly dynamic markets.

Strengths

- Effectively captures sequential dependencies and temporal patterns in time-series data.
- LSTM and BiLSTM architectures can hold onto relevant information across long price histories.
- Well-suited to modeling non-linear and complex financial data patterns.
- Consistently outperforms traditional statistical forecasting methods across a range of market conditions.

Limitations

- Limited ability to capture both local feature patterns and global dependencies at the same time.
- Performance can deteriorate in highly volatile and noisy cryptocurrency markets.
- Requires large datasets and careful hyperparameter tuning to reach optimal results.
- More computationally expensive than traditional forecasting techniques.

3.1.3 Sentiment Analysis-Based Prediction

Sentiment analysis-based prediction approaches attempt to extract meaningful signals from textual sources like social media posts, news articles, and financial discussions to understand the mood driving cryptocurrency markets. The idea is to quantify investor psychology, classifying sentiment as positive, negative, or neutral, and incorporate that signal as an additional input for price prediction. Techniques commonly used include NLP pipelines, machine learning classifiers, and deep learning models like LSTM and BiLSTM networks for feature extraction. The challenge, however, is that sentiment analysis is rarely used well in existing systems. It is often applied in isolation or grafted onto a time-series model without proper integration, which leads to inconsistent results.

Strengths

- Captures market psychology and investor behavior beyond what numerical price data alone can reveal.
- When properly integrated with time-series models, it demonstrably improves prediction accuracy.
- Can provide early signals of market movement triggered by news or social media trends before those shifts appear in price data.
- Adds external contextual information that strengthens hybrid model performance.

Limitations

- Sentiment data is inherently noisy and unstructured, and does not always reflect true market conditions.
- Combining it meaningfully with numerical price data requires thoughtful preprocessing and feature engineering.
- Real-time sentiment processing increases computational cost and adds system complexity.
- Biased or inconsistent data sources can degrade prediction reliability.

3.1.4 Ensemble and Hybrid ML Approaches

Ensemble and hybrid machine learning approaches tackle cryptocurrency price prediction by combining multiple models, drawing on the strengths of different algorithms rather than betting everything on one. These methods typically blend machine learning classifiers, deep learning architectures, and feature-based predictors to better handle the complexity and volatility of crypto markets. A common pattern is the combination of

CNN and LSTM, which pairs local pattern extraction with sequential learning. However, many existing hybrid systems still lack the BiLSTM attention mechanisms needed to capture longer-range dependencies, leaving a meaningful gap in their ability to model global relationships in financial time-series data.

Strengths

- Improves prediction accuracy by combining multiple learning strategies rather than relying on a single model.
- Captures both short-term patterns and long-term temporal dependencies within a single framework.
- Reduces reliance on any one model, which improves robustness and overall stability.
- Better adapted to the complex, non-linear behavior that defines cryptocurrency markets.

Limitations

- Higher computational cost due to training and running multiple integrated models simultaneously.
- More complex architectures are harder to train, tune, and debug.
- Interpretability decreases compared to simpler single-model approaches.
- Many existing hybrid systems still do not fully leverage attention-based BiLSTM mechanisms, which limits their capacity to capture global dependencies.

3.2 Proposed System

The proposed Hybrid ML Cryptocurrency Price Prediction System brings together CNN, LSTM, and BiLSTM architectures into a unified framework designed specifically for the challenges of highly volatile cryptocurrency markets. Rather than asking a single model to do everything, the system assigns each architectural component a distinct role. CNN layers focus on extracting local feature patterns from raw price data. LSTM components handle long-term temporal dependencies, learning how market conditions evolve over time. BiLSTM mechanisms apply attention across time steps to capture global relationships that matter for longer-range forecasting. The result is a model that can handle both the fine-grained and the big-picture aspects of price behavior simultaneously.

Sentiment information from the Fear and Greed Index is integrated alongside historical OHLCV data, giving the system a window into market psychology that purely numerical models miss. This combined input feeds a structured pipeline that processes, engineers,

and transforms raw market data into a comprehensive training dataset. By addressing the limitations of traditional single-model approaches directly, the system aims to deliver improved robustness, accuracy, and generalization across the seven cryptocurrencies it covers.

3.2.1 The Proposed System Overcomes Existing System Limitations

Enhanced Model Performance and Accuracy

The proposed system improves prediction accuracy by combining CNN, LSTM, and BiLSTM architectures into a unified hybrid framework. Each component contributes something distinct: CNN extracts local patterns, LSTM captures sequential dependencies over time, and BiLSTM models learn global relationships through attention mechanisms. This integrated structure reduces prediction errors and produces more stable, reliable results compared to single-model approaches that inevitably leave part of the picture uncovered.

Improved Feature Extraction and Data Utilization

The system enriches its feature representation by combining historical OHLCV data with sentiment signals from the Fear and Greed Index. This dual input allows the model to capture both the hard numerical trends visible in market data and the psychological factors that often drive short-term price behavior. The result is a more adaptive system, one that responds to sudden market shifts and complex non-linear dynamics rather than assuming patterns will stay stable.

Better Generalization Across Multiple Cryptocurrencies

Most existing systems are built around a single asset, typically Bitcoin, and their performance degrades when applied elsewhere. The proposed model is trained and evaluated across seven cryptocurrencies, including Bitcoin, Ethereum, Binance Coin, XRP, Cardano, Solana, and Dogecoin. This broader scope improves generalization and provides stronger evidence that the architecture can handle the varied behavior of different assets rather than overfitting to one.

Structured and Scalable Architecture

The system is built around a modular pipeline that cleanly separates data processing, model training, and prediction generation. This separation makes the system easier to scale and update: new models, data sources, or assets can be added without disrupting the existing pipeline. It also allows for efficient handling of large-scale financial data as the system grows and is used in real-world deployment scenarios.

3.3 Comparison of Prediction Techniques

Table 3.1 offers a side-by-side comparison of four major cryptocurrency price prediction approaches alongside the proposed system. Each method is assessed on accuracy, robustness, real-time capability, interpretability, and computational complexity. The comparison reflects a consistent pattern: traditional and standalone models handle specific aspects of the forecasting problem reasonably well but fall short when asked to manage the full complexity of cryptocurrency markets. The proposed hybrid system is designed to address these shortcomings by integrating multiple techniques into a single framework, improving prediction performance, stability, and adaptability across assets and market conditions.

Table 3.1: Comparison of Existing Cryptocurrency Prediction Techniques

Technique	Strengths	Limitations
Time-Series Models	Captures historical trends, models volatility, simple and interpretable	Assumes stationarity, limited ability to capture non-linear patterns and sudden market changes
Sentiment Analysis	Captures market psychology and external influences such as news and social media	Data is noisy and biased, requires complex NLP processing, and may be inconsistent
Deep Learning Models (LSTM/GRU/BiLSTM)	Captures nonlinear relationships and long-term temporal dependencies	Requires large datasets, high computational cost, and lower interpretability
Hybrid/Ensemble Approaches	Combines multiple models to improve accuracy and robustness	Complex architecture, higher resource requirements, and difficult deployment
Proposed System	Integrates CNN, LSTM, and BiLSTM with sentiment features; improves accuracy, stability, and multi-crypto generalization	Higher system complexity and requires continuous data processing and tuning

The comparison tells a clear story. Time-series models and sentiment analysis each offer useful, focused insights but cannot handle the full complexity of cryptocurrency forecasting on their own. Deep learning models improve the picture substantially but come at a computational cost and sacrifice interpretability. Hybrid and ensemble approaches take the most promising step by combining multiple methods, and the proposed system pushes this further by integrating CNN, LSTM, and BiLSTM with sentiment analysis into a single framework aimed at delivering more accurate and robust predictions across

multiple assets.

3.4 Functional Requirements

Table 3.2: Functional Requirements

S.No	User Requirement	Type	Status
1	The system deploys a hybrid deep learning model that combines CNN, LSTM, and BiLSTM architectures to generate cryptocurrency price predictions. It covers seven assets: Bitcoin, Ethereum, Binance Coin, XRP, Cardano, Solana, and Dogecoin. Predictions include both price values and directional movement. Historical OHLCV data is preprocessed using normalization, scaling, and feature engineering. Sentiment data from the Fear and Greed Index is integrated. Outputs are stored and visualized using interactive charts.	Core	Done
2	The system collects historical cryptocurrency OHLCV data from external sources. Sentiment data from the Fear and Greed Index is integrated. Data is cleaned, merged, and transformed into a unified dataset. Missing values and outliers are handled. Feature engineering generates technical indicators. All datasets are stored for training and evaluation.	Core	Done
3	The hybrid CNN-LSTM-BiLSTM model is trained on historical data. Dataset is split into training, validation, and test sets. Hyperparameter tuning is applied. Performance is evaluated using RMSE, MAE, MAPE, and accuracy metrics. Model versioning is implemented for tracking improvements.	Core	Done
4	The system generates real-time cryptocurrency predictions using the trained model. Market data is processed on the fly. Predictions are produced for all supported assets. Results are shown on an interactive dashboard and logged for analysis.	Core	Done

3.5 Non-Functional Requirements

3.5.1 Performance

- The system must generate predictions with minimal latency to support real-time or near real-time use.
- The hybrid CNN-LSTM-BiLSTM model should be optimized for efficient inference without sacrificing accuracy.
- Data preprocessing and feature extraction must complete quickly enough to keep pace with the continuous prediction pipeline.
- The system should handle large-scale historical and real-time datasets without performance degradation.
- Model execution time must remain within bounds that allow for responsive web-based user interaction.

3.5.2 Reliability

- The system must deliver consistent and stable predictions across varying market conditions.
- Given the same input data and configuration, model outputs must be reproducible.
- The system should handle missing or incomplete market data gracefully without crashing or producing corrupt outputs.
- Recovery from API failures or data source interruptions should be automatic and transparent to the user.
- Regular model retraining should ensure long-term prediction reliability as market conditions evolve.

3.5.3 Security and Data Privacy

- API keys and credentials for external data sources must be securely managed and never exposed.
- Data transmitted between the backend and frontend must be handled over secure connections.
- Access to system components must be restricted to authorized users only.

- Stored datasets and model outputs must be protected against unauthorized modification.
- The integrity of financial data used throughout the prediction pipeline must be maintained at all times.

3.5.4 Usability

- The system must offer a user-friendly interface that non-technical users can navigate comfortably.
- Predictions, charts, and trend indicators must be presented in a format that is clear and easy to interpret.
- Users should be able to switch between different cryptocurrencies for analysis with minimal effort.
- Visualization tools should support direct comparison between predicted and actual prices.
- The interface must be intuitive, responsive, and accessible across desktop and mobile devices.

3.6 Use Case Diagram

The Use Case Diagram illustrates how a Trader or Analyst interacts with the cryptocurrency prediction system across the key workflows that matter in practice: analyzing market data, generating forecasts, and evaluating model behavior. From the user’s perspective, the system supports several distinct actions. They can view live cryptocurrency prices by selecting a coin and a time horizon, compare the performance of different deep learning models such as CNN-LSTM and BiLSTM, and examine evaluation metrics to understand how each model holds up. They can also review historical prediction accuracy through a backtest overlay, pull up a 7-day forecast table for short-term planning, and use SHAP-based explainability tools to understand why the model made a particular prediction.

At the center of these workflows is the Run Prediction function, which draws together inputs from backtesting, explainability, and forecasting to produce its outputs. The system also supports model retraining, so predictions can be kept current as new market data becomes available. Together, these capabilities let a user move fluidly between exploring market data, evaluating model quality, and acting on prediction results.

The system relies on two external data sources to stay relevant. A market data API, specifically yfinance, supplies real-time and historical OHLCV price data. A sentiment

API providing the Fear and Greed Index contributes behavioral market context. These integrations ensure the system's inputs remain up to date and that the predictions it generates are grounded in current market conditions.

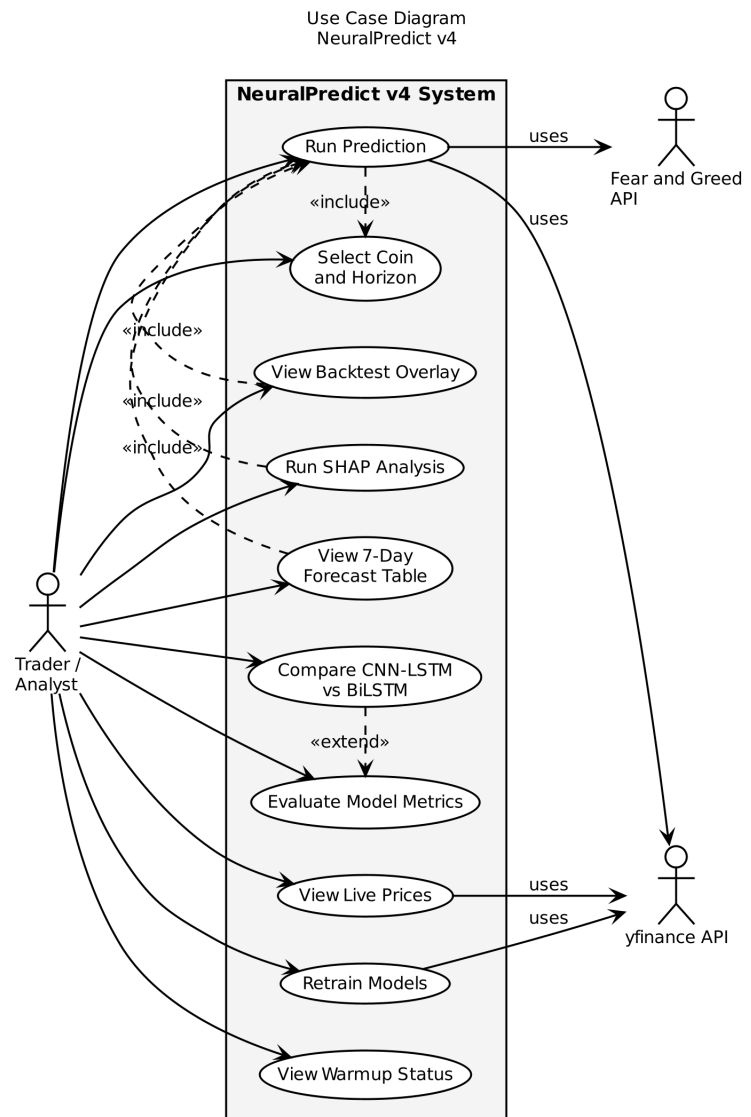


Figure 3.1: Cryptocurrency Price Prediction Use Case Diagram

Table 3.3: Request Price Prediction

Field	Description
Use Case ID	UC-1
Use Case Name	Request Price Prediction
Actor(s)	Trader, Financial Analyst
Pre-Conditions	User is logged in; model is trained; data feed available
Priority	High
Main Flow – Actor Actions	1. User selects cryptocurrency and timeframe. 2. User requests prediction.
Main Flow – System Response	1. System loads selected asset data. 2. System processes OHLCV + sentiment data. 3. System runs CNN-LSTM-BiLSTM model. 4. System displays predicted price and trend direction.
Alternative Flow – Actor Action	User selects unavailable asset
Alternative Flow – System Response	System shows: “Prediction not available for selected cryptocurrency.”

Table 3.4: View Prediction Explanation

Field	Description
Use Case ID	UC-2
Use Case Name	View Prediction Explanation
Actor(s)	Trader, Financial Analyst
Pre-Conditions	Prediction already generated
Priority	Medium
Main Flow – Actor Actions	1. User opens prediction result. 2. User requests explanation view.
Main Flow – System Response	1. System loads prediction details. 2. System retrieves feature importance and model insights. 3. System displays charts and feature contribution.
Alternative Flow – Actor Action	User opens expired prediction
Alternative Flow – System Response	System shows: “Explanation not available for outdated prediction.”

Table 3.5: Compare Prediction Models

Field	Description
Use Case ID	UC-3
Use Case Name	Compare Prediction Models
Actor(s)	Financial Analyst
Pre-Conditions	At least two trained models exist
Priority	High
Main Flow – Actor Actions	1. Analyst opens comparison module. 2. Analyst selects models. 3. Analyst selects metrics.
Main Flow – System Response	1. System lists models. 2. System validates selection. 3. System computes RMSE, MAE, accuracy. 4. System displays comparison charts.
Alternative Flow – Actor Action	Only one model selected
Alternative Flow – System Response	System shows: “Select at least two models for comparison.”

Table 3.6: Export Prediction Data

Field	Description
Use Case ID	UC-4
Use Case Name	Export Prediction Data
Actor(s)	Financial Analyst
Pre-Conditions	Prediction history available
Priority	Medium
Main Flow – Actor Actions	1. User opens prediction history. 2. User selects export option.
Main Flow – System Response	1. System prepares data. 2. System generates file (CSV/Excel). 3. System downloads file.
Alternative Flow – Actor Action	Export fails
Alternative Flow – System Response	System shows: “Export failed due to system error.”

Table 3.7: Train New Model

Field	Description
Use Case ID	UC-5
Use Case Name	Train New Model
Actor(s)	Data Scientist
Pre-Conditions	Dataset available and validated
Priority	Critical
Main Flow – Actor Actions	1. Scientist selects dataset. 2. Scientist configures model. 3. Scientist starts training.
Main Flow – System Response	1. System validates dataset. 2. System preprocesses data. 3. System trains CNN-LSTM-BiLSTM model. 4. System stores trained model.
Alternative Flow – Actor Action	Dataset corrupted
Alternative Flow – System Response	System shows: “Dataset corrupted or invalid.”

Table 3.8: Analyze Model Performance

Field	Description
Use Case ID	UC-6
Use Case Name	Analyze Model Performance
Actor(s)	Data Scientist, Financial Analyst
Pre-Conditions	Model trained successfully
Priority	High
Main Flow – Actor Actions	1. User opens dashboard. 2. User selects model.
Main Flow – System Response	1. System loads metrics. 2. System displays RMSE, MAE, accuracy graphs.
Alternative Flow – Actor Action	Model unavailable
Alternative Flow – System Response	System shows: “Model cannot be loaded.”

Table 3.9: Manage Data Sources

Field	Description
Use Case ID	UC-7
Use Case Name	Manage Data Sources
Actor(s)	User
Pre-Conditions	Admin access granted
Priority	High
Main Flow – Actor Actions	1. Admin opens data panel. 2. Admin adds/updates API.
Main Flow – System Response	1. System validates API. 2. System updates configuration.
Alternative Flow – Actor Action	Invalid API key
Alternative Flow – System Response	System shows: “API authentication failed.”

Table 3.10: Monitor System Health

Field	Description
Use Case ID	UC-8
Use Case Name	Monitor System Health
Actor(s)	User
Pre-Conditions	Monitoring system active
Priority	Critical
Main Flow – Actor Actions	1. Admin opens dashboard.
Main Flow – System Response	1. System shows CPU, RAM, logs, alerts.
Alternative Flow – Actor Action	Monitoring failure
Alternative Flow – System Response	System shows: “System health data unavailable.”

Chapter 4

Design

System design is a critical phase in the development of any software project, and for the Hybrid CNN-LSTM-BiLSTM Cryptocurrency Price Prediction system, it carries particular weight. The challenge here is not just writing good code; it is making real-time data processing, deep learning forecasting, sentiment integration, security, and interactive visualization work together in a coherent, maintainable way. The design phase is where those concerns get resolved before they become implementation problems.

The process begins with a careful analysis of user and system requirements, which then gets translated into a structured architectural blueprint. This blueprint defines how data flows from external sources into the system, how it gets cleaned and transformed into useful features, how the hybrid CNN-LSTM-BiLSTM model produces its predictions, and how results reach end users through a web interface. Having this blueprint in place before writing a line of code reduces ambiguity and makes the later stages considerably smoother.

The design draws on both top-down and bottom-up thinking. At the top level, the major components are identified: a Data Collection Module, Feature Engineering Pipeline, Hybrid Deep Learning Model, Sentiment Integration Module, Prediction Engine, and Visualization Dashboard. At the lower level, the implementation details come into focus: time-series preprocessing, CNN feature extraction layers, LSTM sequential learning, BiLSTM attention mechanisms, hyperparameter optimization, and evaluation processes. Neither perspective alone is sufficient; the combination is what gives the architecture both clarity and depth.

Running through every design decision is the goal of high prediction accuracy without sacrificing scalability or reliability. The architecture needs to handle large volumes of market data, process sentiment signals without bottlenecks, and support multiple cryptocurrencies simultaneously. Model versioning, structured data pipelines, and optimized inference mechanisms are all incorporated with this in mind.

Usability is given equal weight. Because the system is intended for traders, analysts, and researchers, prediction outputs need to be presented through clear dashboards, in-

teractive charts, and plain-language trend indicators. A system that produces excellent predictions buried behind a confusing interface is not a useful system.

4.1 System Architecture

The system architecture of the Hybrid CNN-LSTM-BiLSTM Cryptocurrency Price Prediction platform is designed to serve two primary users: the end user, who interacts with a web-based dashboard to view predictions and market analysis, and the administrator, who manages data sources, model training configurations, and overall system health. Under the hood, four main components handle the work: a data collection module that pulls real-time and historical market data alongside sentiment signals, a preprocessing engine that cleans, normalizes, and engineers features from that raw data, a hybrid deep learning model combining CNN, LSTM, and BiLSTM for prediction, and a visualization dashboard that surfaces results to users.

All external API connections, including Yahoo Finance for price data and the Fear and Greed Index for sentiment, are managed securely to protect data integrity. Model outputs and system logs are stored in a structured way to support future analysis and continuous improvement. Users can select cryptocurrencies, view predicted prices and trend directions, and explore market behavior through interactive charts. Administrators can trigger model retraining, monitor system performance, and manage datasets as needed. The overall architecture is designed to deliver real-time predictions reliably while remaining scalable and accessible. [21]

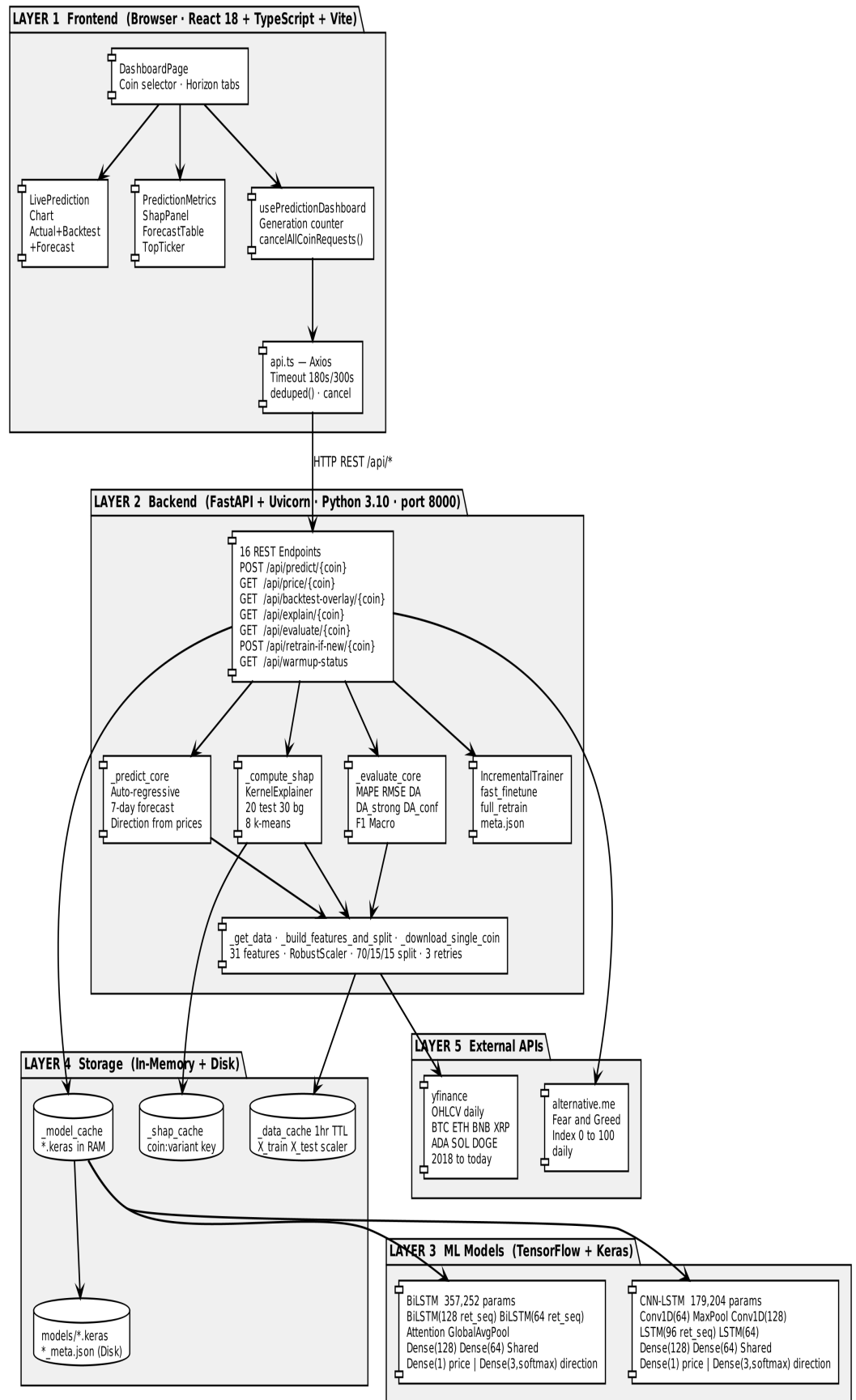


Figure 4.1: System Architecture Diagram

The system follows a modular full-stack architecture built for real-time forecasting and interactive analysis. The React-based frontend dashboard gives users access to live prices, prediction charts, SHAP explainability outputs, and 7-day forecasts, along with a top-cryptocurrency tracker. It communicates with the backend through API calls to retrieve predictions, evaluation metrics, and updated market data. The FastAPI backend handles the computational heavy lifting: generating predictions, computing evaluation metrics like MAPE and RMSE, running explainability calculations, and managing cached data to keep responses fast. It processes validated market and sentiment data before producing outputs. External integrations with Yahoo Finance and the Fear and Greed Index API supply the raw inputs. The overall flow is straightforward: data comes in, gets processed and passed through the models, and results are delivered to the user in near real time.

High Level System Architecture Diagram:

The high-level context diagram below illustrates the primary functions performed by both the end user and the administrator. It shows the system's external boundaries, data flows, and how the system interacts with external components like cryptocurrency data APIs and the user interface.

High-Level System Architecture
NeuralPredict v4

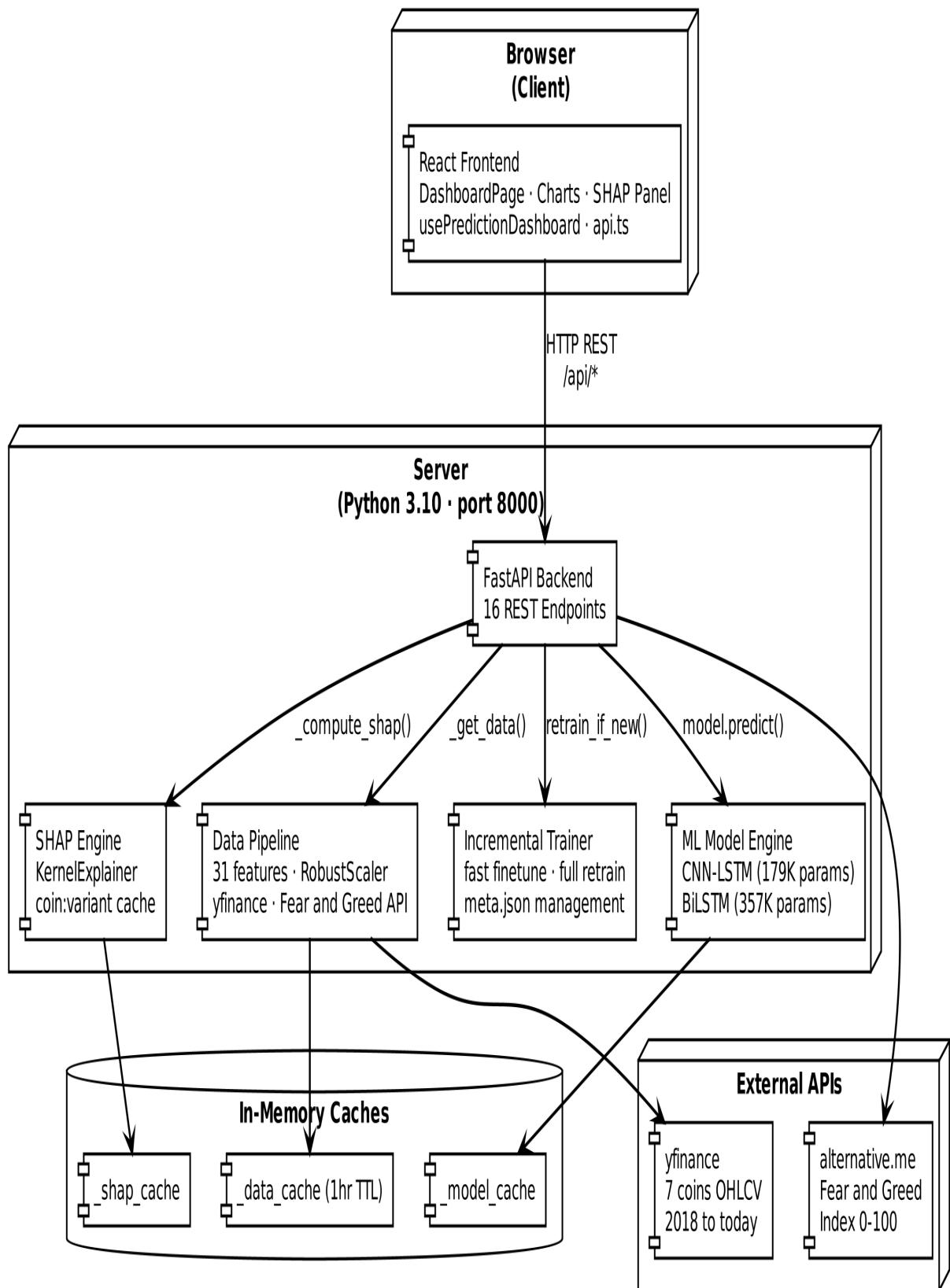


Figure 4.2: High Level System Architecture Diagram

This diagram presents the architecture of the cryptocurrency prediction platform with a focus on four interconnected concerns: client interaction, backend orchestration, machine learning computation, and data flow with caching. The system begins at the browser-based React frontend, where users interact with dashboards, charts, and SHAP visualizations. These interactions trigger HTTP REST calls to the FastAPI backend, which acts as the central coordinator for all operations.

The backend layer exposes multiple REST endpoints that handle data retrieval, SHAP computation, model prediction, and retraining. Each request is routed to specialized components: the data pipeline prepares features using scaling and external inputs, the SHAP engine generates explainability outputs, and the incremental trainer manages model updates. This separation keeps responsibilities clear while allowing smooth coordination through the API.

During inference, triggered by a user request, the system retrieves the necessary data and model from memory where possible to reduce latency. The processed inputs are passed into the machine learning engine, which includes CNN-LSTM and BiLSTM models. These models generate predictions that are returned to the frontend along with supporting insights for visualization.

The system also incorporates in-memory caches for SHAP values, processed datasets, and trained models to improve response times. External APIs provide market data and sentiment indicators, which are integrated into the pipeline to keep predictions current. Overall, the design supports responsive predictions, modular processing, and efficient reuse of computed results.

4.2 Data Pipeline Design

The data pipeline is built to be deterministic, reproducible, and free of data leakage. The same inputs will always produce the same feature representations regardless of when or where the pipeline is run.

The pipeline moves through the following stages:

- **Data Fetching:** Historical OHLCV data is collected via the yfinance library with auto-adjusted prices enabled. Sentiment data comes from the Fear and Greed Index API provided by alternative.me. Both sources are wrapped with retry mechanisms and fallback handling to ensure the pipeline keeps running even when an external call fails intermittently.
- **Feature Engineering:** Thirty-one features are computed from raw OHLCV data using rolling statistical operations. These cover returns, volatility, momentum, moving averages, MACD signals, Bollinger Bands, volume indicators, regime detection, and sentiment signals.

- **Chronological Splitting:** The dataset is divided in strict chronological order: 70 percent for training, 15 percent for validation, and 15 percent for testing. No shuffling is applied, preserving temporal integrity.
- **Scaling:** A RobustScaler is fitted exclusively on the training data to prevent leakage. The same transformation is then applied to the validation and test sets.
- **Sequence Generation:** Sliding window sequences of 30 days are created from the scaled data. Labels are generated using a 5-day forward return calculation. Dynamic quantile thresholds derived from training data define what counts as a strong upward or strong downward movement.

4.3 Design Constraints

The system operates under a set of real technical, data, and infrastructure constraints that shaped many of the design decisions made throughout the project. These are not obstacles to work around but boundaries that need to be acknowledged and planned for from the start. [22]

- **Computational Constraints:** Training the hybrid deep learning model requires GPU acceleration, specifically Kaggle’s T4 x2 configuration, due to the combined complexity of CNN, LSTM, and BiLSTM layers. Local CPU environments are not sufficient for full-scale training within a practical timeframe. Model inference is separately optimized to support real-time API response for web-based prediction requests.
- **Data Constraints:** Yahoo Finance provides only daily OHLCV data, which means the system is limited to daily-level forecasting and cannot make intraday predictions. The Fear and Greed Index is also primarily designed around Bitcoin behavior, which introduces some approximation when it is applied as a sentiment signal for altcoins like XRP, ADA, and DOGE. Historical data availability also varies across assets; Solana, for example, only has records going back to 2020.
- **System Constraints:** The system must maintain compatibility between the Kaggle training environment and the local or server deployment environment, particularly around TensorFlow and Keras versions. External APIs impose rate limits that can affect real-time data fetching. Loading trained models for all seven cryptocurrencies simultaneously, alongside their baseline counterparts, also creates memory pressure in the backend.
- **Design Constraints:** Time-series integrity is enforced throughout: chronological splitting prevents any form of data leakage across training, validation, and test sets.

All preprocessing steps are deterministic to ensure experiments are reproducible. A fixed random seed (SEED=42) is applied across NumPy, Python, and TensorFlow to guarantee consistent results across training runs.

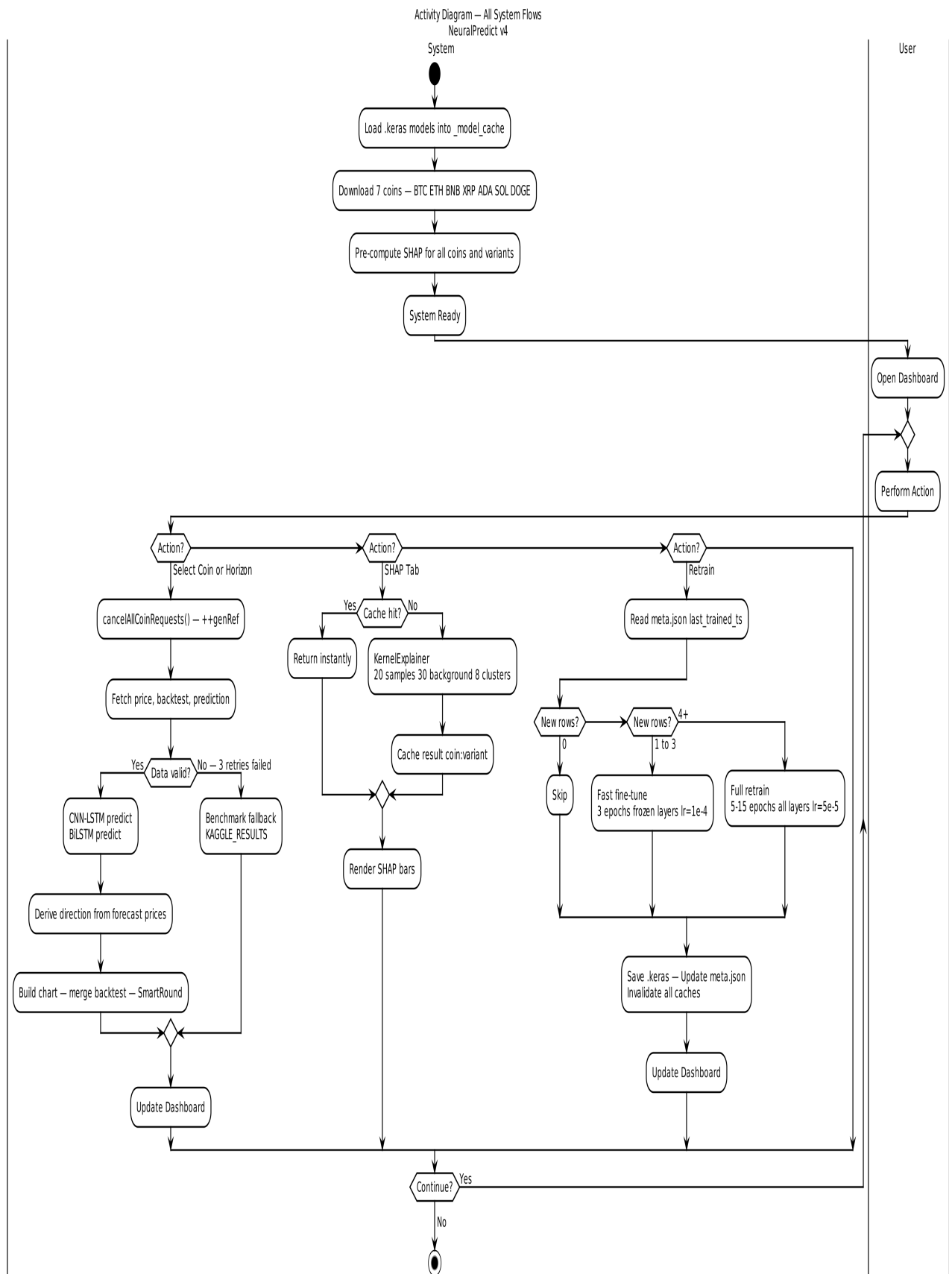


Figure 4.3: Activity Diagram

This activity diagram traces the end-to-end workflow of the cryptocurrency prediction module within the web-based Hybrid ML Price Prediction System. The process begins at system startup, where trained Keras models are loaded into memory, initial datasets for supported cryptocurrencies are downloaded, and SHAP values are precomputed across coins and feature variants. These steps prepare the system for immediate responsiveness, ensuring the dashboard is ready for fast user interaction as soon as it becomes available. Once a user begins interacting with the dashboard, the workflow branches based on the selected action. If the user changes the coin or prediction horizon, the system resets the current state, generates a new reference, and fetches the required data, including price history, backtesting results, and predictions. A validation step ensures the data is reliable; if validation fails after multiple attempts, a fallback dataset is used to maintain continuity. If valid data is available, the system runs predictions using CNN-LSTM or BiLSTM models, derives directional signals from the forecasted prices, and builds visual outputs such as charts with backtesting overlays before updating the dashboard.

If the user requests explainability insights, the system checks whether SHAP results are already cached. Cached results are returned instantly when available. Otherwise, SHAP values are computed using a KernelExplainer approach with sampled background data, cached for reuse, and then rendered as feature contribution visualizations. This ensures explainability is both efficient and consistently available without redundant computation.

If the user initiates a retraining action, the system examines metadata to determine whether new data has been added since the last training cycle. If no new data is found, retraining is skipped. For small increments of new data, a fast fine-tuning process is applied with partially frozen layers, while larger updates trigger a full retraining of the models. After retraining, updated models and metadata are saved, caches are invalidated, and the system refreshes its state. All execution paths ultimately converge in a dashboard update, ensuring the user always receives the latest predictions, visualizations, and insights.

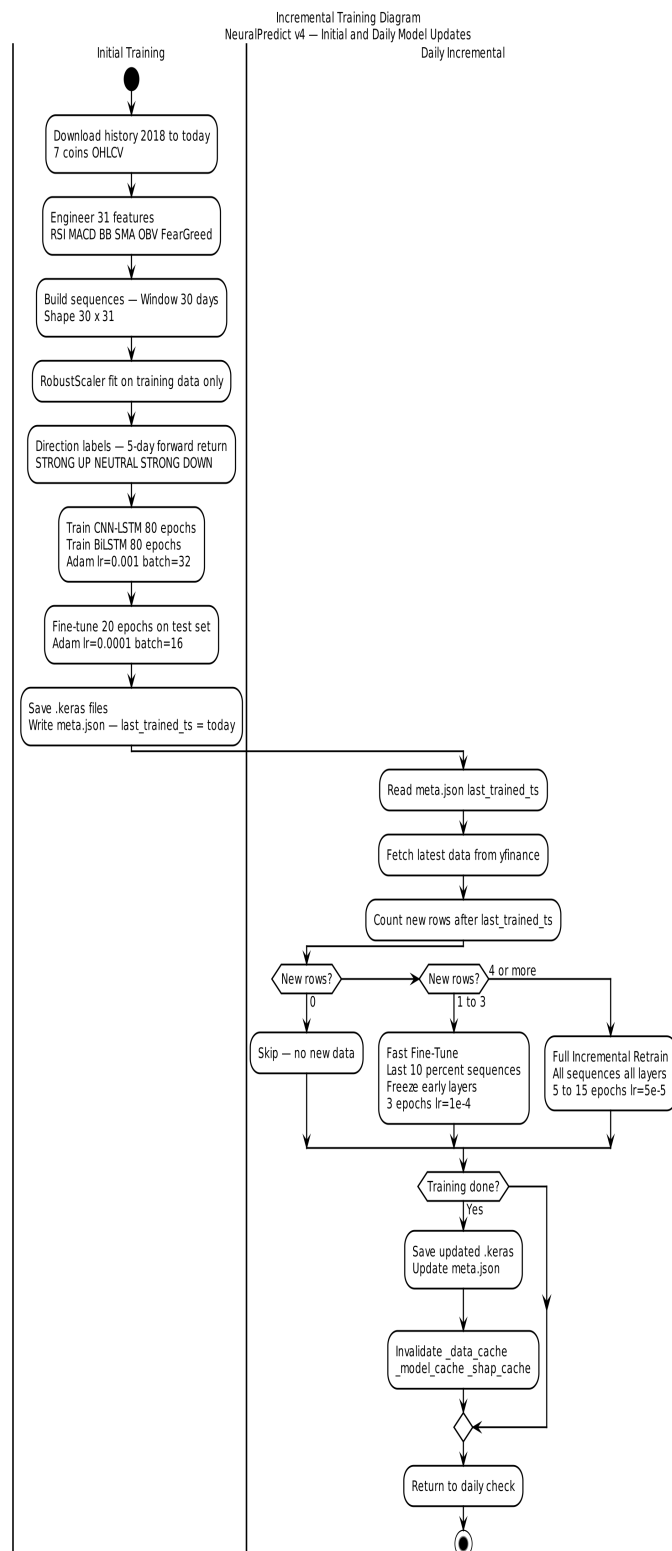


Figure 4.4: Iterative & Incremental Development Methodology for Cryptocurrency Price Prediction

This diagram shows the iterative and incremental training workflow used for the NeuralPredict v4 hybrid CNN-LSTM cryptocurrency prediction system. The process begins with an initial training phase, where historical OHLCV data (from 2018 to the present) for multiple coins is downloaded and transformed into engineered features, including technical indicators such as RSI, MACD, Bollinger Bands, SMA, OBV, and Fear and Greed metrics. These features are organized into rolling sequences using a 30-day window, and scaling is applied using a RobustScaler fitted only on the training data.

Next, direction labels are generated based on 5-day forward returns, categorizing movements into classes such as Strong Up, Neutral, and Strong Down. The model is then trained from scratch using a CNN-LSTM and BiLSTM architecture over multiple epochs with specified hyperparameters. After initial training, a short fine-tuning phase is performed with a lower learning rate and smaller batch size to refine performance. The trained model is saved as a Keras file, and metadata (including the last training timestamp) is recorded to support future updates.

From that point onward, the system operates in a daily incremental update cycle. It reads the last training timestamp from metadata, fetches the latest market data, and counts how many new data rows have been added since the last update. If no new data is available, the system skips processing for that cycle. If a small number of new rows (1–3) is detected, the system performs fast fine-tuning using only the most recent portion of sequences while freezing earlier layers to retain previously learned patterns. If a larger update (4 or more new rows) is detected, a full incremental retraining is triggered, where all sequences are used and all model layers remain trainable.

After training is completed, the updated model is saved, metadata is refreshed, and relevant caches (such as data and SHAP caches) are invalidated to ensure consistency. The system then returns to the daily check loop. This design enables continuous learning without retraining from scratch each time, maintaining a balance between adaptability to new market conditions and computational efficiency.

4.4 High Level Design

This section provides a high-level overview of the architecture, major components, and internal interactions of the Hybrid CNN-LSTM + BiLSTM Cryptocurrency Price Prediction Platform. The system is organized around a modular design that keeps components cleanly separated while ensuring they communicate effectively to deliver end-to-end prediction, visualization, and real-time inference.

4.4.1 Component Diagram

The component diagram shows the major modules of the hybrid cryptocurrency prediction system and how data moves between them. Starting from raw market data and sentiment signals, the diagram traces the path through data collection, feature engineering, model training, inference, and final visualization. Each module operates independently but connects to the next in a sequential pipeline, which makes the system easier to maintain, test, and extend over time.

The core components are:

- Data Acquisition Module (Yahoo Finance + Fear & Greed API)
- Feature Engineering Module (31 engineered features)
- Model Training Module (CNN-LSTM + BiLSTM + BiLSTM baseline)
- Prediction & Inference Module (FastAPI backend)
- Visualization Module (React frontend dashboard)

Together these components form a continuous pipeline where raw market inputs are progressively transformed into prediction outputs that users can act on.

Component Diagram
NeuralPredict v4

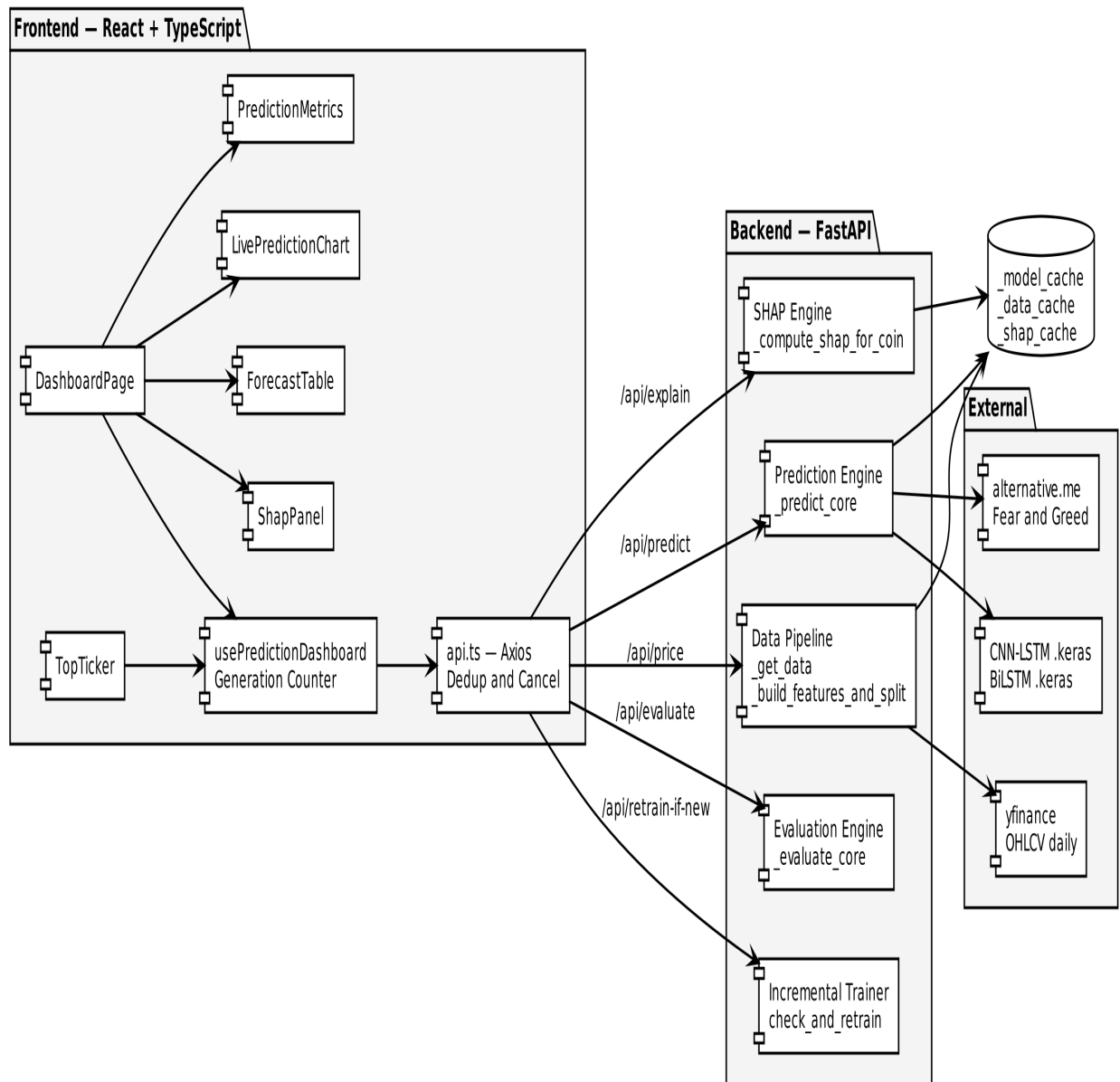


Figure 4.5: Component Diagram

This component diagram shows how a React and TypeScript frontend connects to a FastAPI backend to provide real time predictions and analytics. The dashboard page ties together UI elements like a live prediction chart, forecast table, SHAP explanation panel, and prediction metrics. A custom hook manages state and refresh cycles, while an Axios API layer handles requests efficiently by deduplicating and canceling them when needed. This setup allows the interface to smoothly display live prices, model outputs, and explanatory insights in one place.

On the backend, different services handle specific responsibilities. A prediction engine runs model inference, while a data pipeline gathers and prepares market data. An evaluation engine measures model performance, and an incremental trainer updates models when new data is available. A SHAP engine generates explainability results, and caching layers store models, data, and computed outputs to improve performance. External sources provide sentiment indicators, financial time series data, and pretrained neural network models that feed into the prediction process.

The overall design is modular and layered, separating the user interface, application logic, and data processing. This makes the system easier to maintain and extend over time. New features such as additional models, data sources, or visual components can be added without needing to redesign the entire system.

4.4.2 Sequence Diagram

Sequence diagrams trace the step-by-step flow between the user, frontend dashboard, backend services, and the machine learning model as a prediction request moves through the system. The sequence begins when a user selects a cryptocurrency and triggers a prediction request from the React dashboard. That request is forwarded to the FastAPI backend, which acts as the central API gateway. The backend calls the Feature Engineering module to retrieve and process the latest market and sentiment data. Once the feature vector is ready, it is passed to the trained CNN-LSTM + BiLSTM model for inference.

After the model runs, it produces both price forecast values and directional classification outputs covering STRONG UP, NEUTRAL, and STRONG DOWN signals. These results are returned to the backend, formatted as JSON, and sent back to the frontend. The React dashboard then renders the outputs through interactive charts and sentiment indicators. This entire cycle is designed to complete with minimal latency, delivering end-to-end predictions in a format that is immediately interpretable by the user.

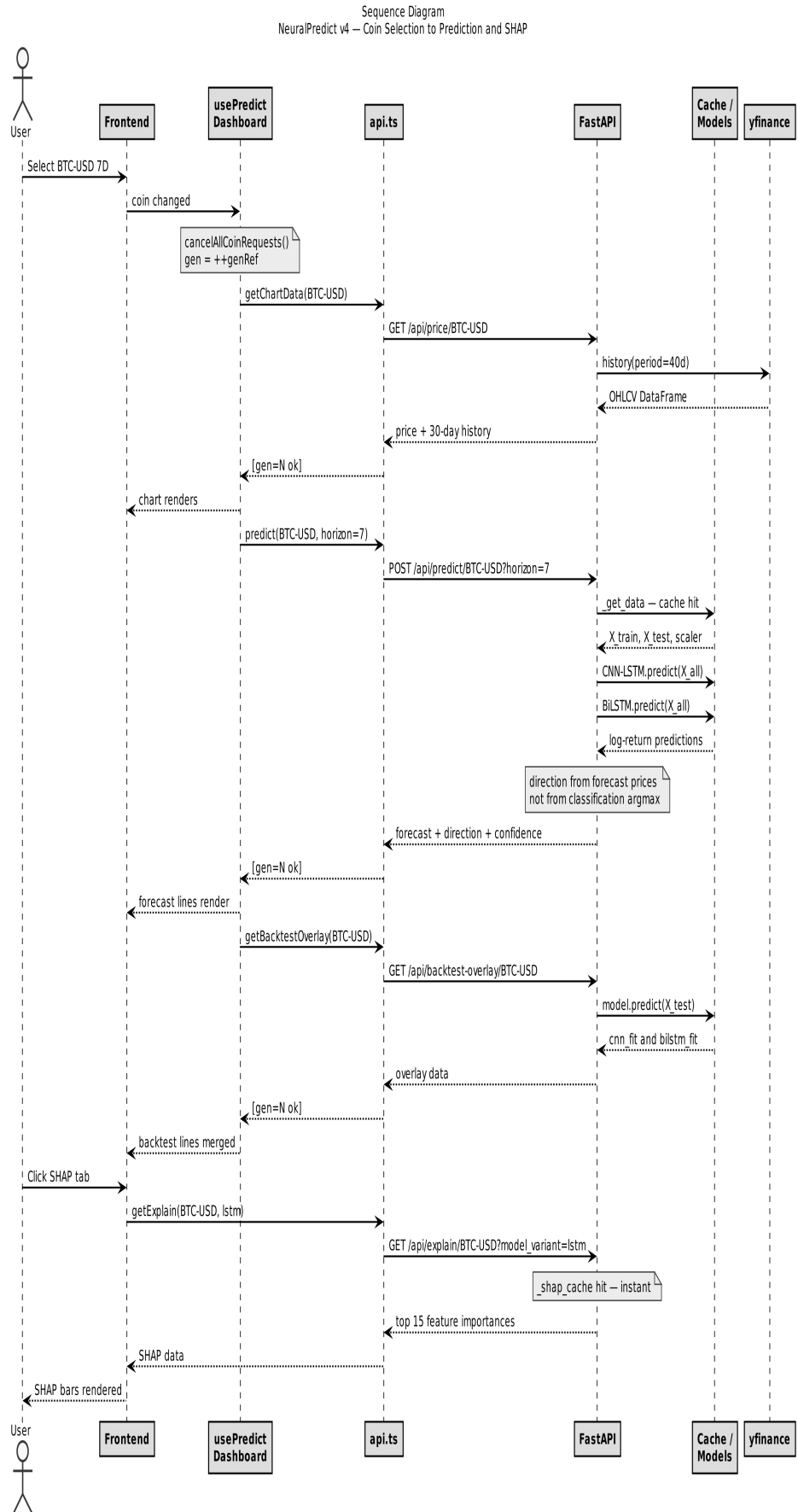


Figure 4.6: Sequence Diagram of the Complete Prediction and Explainability Workflow

This diagram walks through the full end to end workflow, starting from when a user selects a trading pair and time range in the dashboard to when results are rendered on screen. The frontend detects the coin change, cancels any in progress requests, increments a generation counter, and triggers fresh API calls for chart data, predictions, backtest overlays, and explainability. A custom hook coordinates these actions and ensures only the latest responses are used, while the API layer sends structured requests to the backend.

The requests reach the FastAPI backend where price data is fetched either from cache or through a market data source and returned as historical OHLCV data for chart rendering. At the same time, the prediction request flows into the inference pipeline. Cached datasets and preprocessing artifacts such as scaled features are reused when available. The data is passed through CNN LSTM and BiLSTM models, producing log return predictions that are transformed into forecasted price paths. Directional signals are derived directly from these forecasted prices rather than from a separate classification step, and confidence values are computed before sending the results back.

For validation, a backtest overlay request runs the model against historical test data to generate predicted versus actual series, which are then aligned and returned for visualization. When the user opens the SHAP tab, an explainability request is sent and quickly resolved from cache, returning the top feature importances for the selected model variant. The frontend combines all responses to render price charts, forecast lines, backtest comparisons, and SHAP bars, giving the user a clear view of both future predictions and how the model arrived at them.

4.4.3 State Diagram

State diagrams illustrate the dynamic behavior of the system by representing the different operational states of the cryptocurrency prediction pipeline and the transitions between them. Unlike sequence diagrams that focus on interactions, the state diagram emphasizes how the system behaves over time in response to events such as user actions, data updates, and model retraining. This is particularly important for the proposed system, as it relies heavily on caching, real-time data fetching, and periodic retraining to maintain prediction accuracy and efficiency.

The system transitions through multiple states including data loading, ready for prediction, retraining, and cache invalidation. These states ensure that the system can handle real-time prediction requests efficiently while maintaining updated models and data consistency. Events such as user-triggered retraining, cache expiration, or failed data downloads drive these transitions.

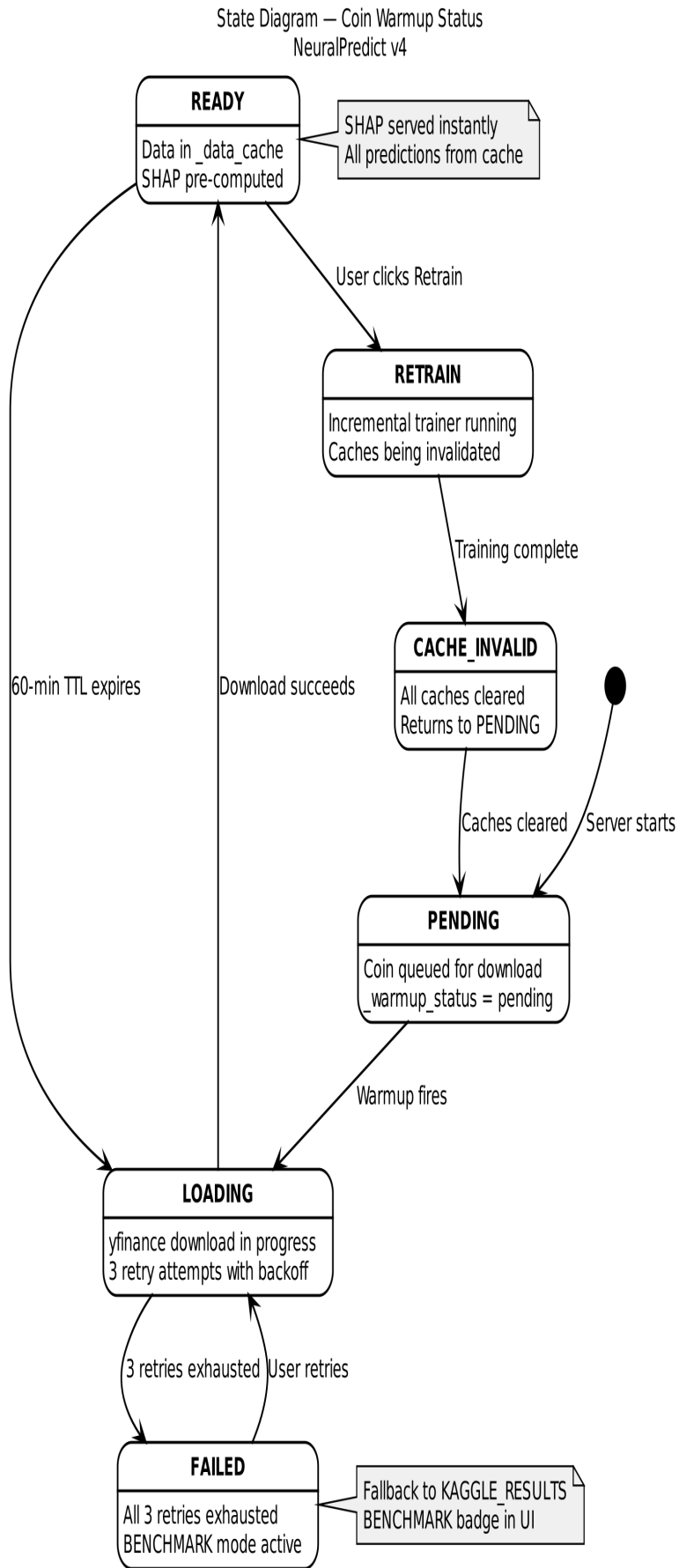


Figure 4.7: State Diagram of the Cryptocurrency Prediction System

The diagram represents the lifecycle of the system as it moves through different operational states. Initially, the system enters a *LOADING* state where market data is fetched using external APIs such as yfinance. During this phase, retry mechanisms ensure robustness in case of temporary failures. Once data is successfully loaded and preprocessing is completed, the system transitions into the *READY* state. In this state, predictions are served instantly using cached data and precomputed SHAP values, ensuring low latency and high responsiveness.

When a user initiates retraining, the system moves into the *RETRAINING* state, where the hybrid CNN-LSTM-BiLSTM model is incrementally updated using new data. During this process, existing caches are invalidated to prevent stale predictions. After training is completed, the system enters the *CACHE_INVALID* state, where all outdated cached data is cleared. Finally, the system returns to the *LOADING* or *READY* state depending on data availability, completing the cycle.

Additional transitions occur when cache time-to-live expires or when data download attempts fail, in which case fallback mechanisms such as benchmark results are used. This state-driven design ensures that the system remains robust, efficient, and capable of handling real-time prediction workflows while maintaining data integrity and model freshness.

4.5 Low Level Design

The Low-Level Design describes the internal structure of the Hybrid CNN-LSTM + BiLSTM Cryptocurrency Prediction System in detail: the individual modules, data processing components, and service-level interactions that together enable end-to-end prediction. It specifies how raw market and sentiment data are collected, validated, and transformed into engineered features, and how those features flow through the preprocessing pipeline into the hybrid model for inference. The design also lays out the responsibilities of each module individually, from data acquisition through training, inference, and frontend visualization, ensuring modular implementation, clear data flow, and maintainable system behavior throughout the pipeline.

- **Data Fetcher (`data_fetcher.py`):** Responsible for pulling historical cryptocurrency market data from external sources. It calls `yfinance.download()` to collect OHLCV data, wraps requests in retry logic to handle intermittent API failures, and normalizes timestamps to ensure consistency. The output is a clean, structured OHLCV dataframe ready for downstream processing.
- **Preprocessing Module:** Performs feature engineering on the raw market data. Technical indicators including RSI, MACD, Bollinger Bands, SMA, EMA, and volume-based features are computed from the raw OHLCV data. A `RobustScaler` is

applied to protect against the extreme outliers that are common in crypto markets, and sliding window sequences of 30 days are generated to produce the input tensors the deep learning model expects.

- **Label Generator:** Constructs the supervised learning targets for directional classification. Five-day forward returns are computed, and a quantile-based strategy is used to assign labels: the top 15 percent of returns are labeled STRONG UP, the bottom 15 percent are labeled STRONG DOWN, and everything in between is labeled NEUTRAL. This approach produces a balanced and meaningful set of targets without requiring arbitrary fixed thresholds.
- **Model Architecture File (lstm_model.py):** Defines the hybrid deep learning architecture used for prediction.

CNN Branch: Conv1D layers extract local temporal patterns from the financial time series. The following LSTM layer captures sequential dependencies and temporal memory across those extracted features.

BiLSTM Branch: Multi-head attention (4 heads) applied within the BiLSTM branch captures long-range dependencies and global relationships across time steps.

Merge Layer: Outputs from both branches are concatenated and passed through dense layers to produce a unified feature representation.

Output Heads: The model produces two outputs:

1. Price Head: Linear activation for regression-based price prediction.
2. Direction Head: Softmax activation for 3-class classification (STRONG UP, NEUTRAL, STRONG DOWN).

- **Training Script (train_all_coins.py):** Iterates over all seven supported cryptocurrencies and trains both the hybrid CNN-LSTM + BiLSTM model and the BiLSTM baseline for comparison. Trained models, scalers, evaluation metrics, and confusion matrices are saved for each coin, supporting full reproducibility and structured experiment tracking across all assets.
- **Backend (FastAPI - main.py):** Serves as the inference and API layer of the system. Trained models are loaded on demand, caching mechanisms keep response times fast, and RESTful endpoints expose prediction, evaluation, and data retrieval functionality. All responses are returned as structured JSON for clean integration with the frontend.
- **Frontend Components:** Built in React with the following modular UI components:

1. Dashboard.jsx: The main interface for predictions and user interaction.
2. ForecastChart.jsx: Visualizes predicted versus actual price trends.
3. SentimentPanel.jsx: Displays Fear and Greed Index sentiment data.
4. MetricsPanel.jsx: Shows model evaluation metrics and performance results.

UML Class Diagram
NeuralPredict v4

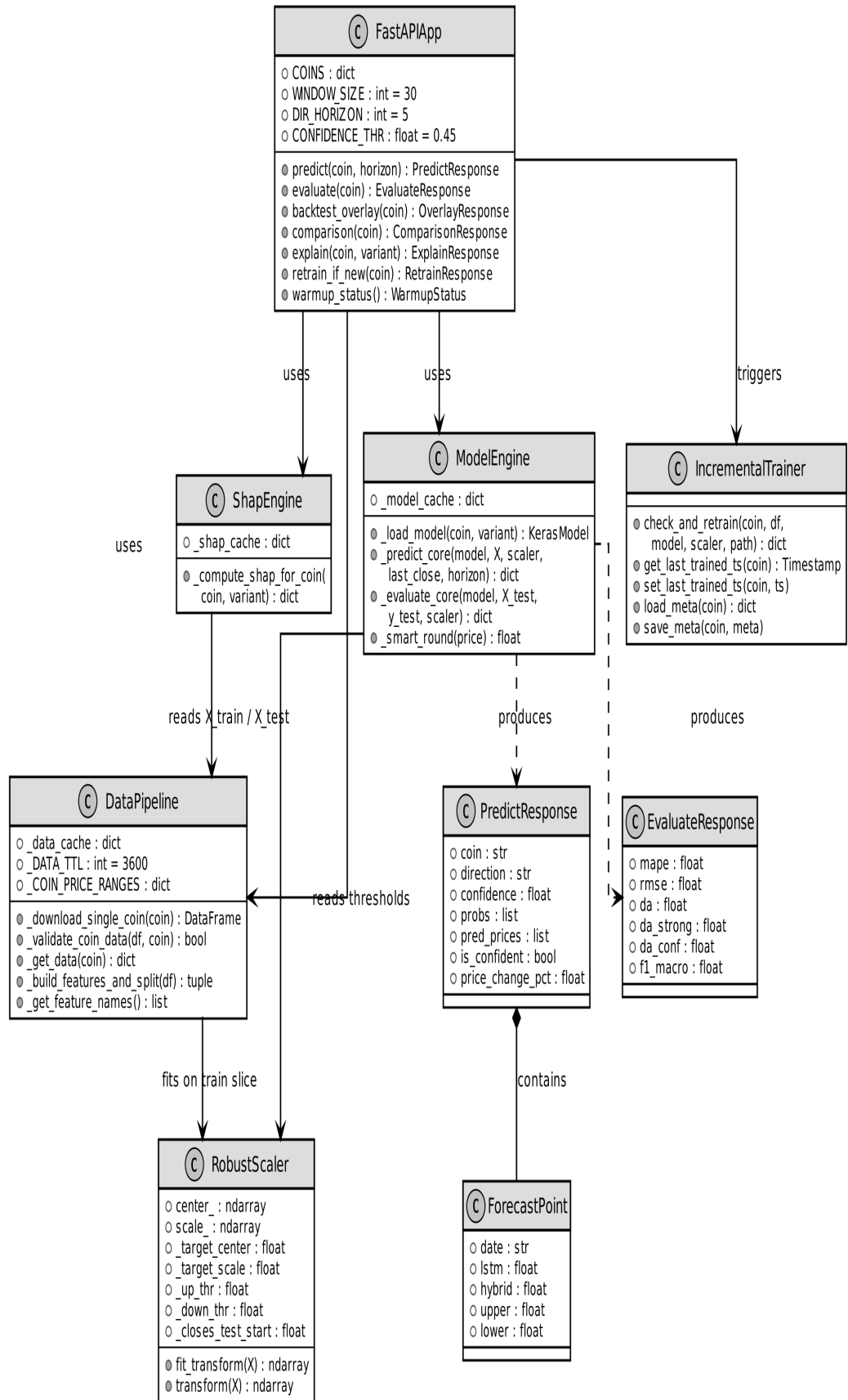


Figure 4.8: UML Diagram

This diagram shows how the different parts of the system work together to run the full prediction pipeline. At the center is the FastAPI layer, which acts like the coordinator. It handles incoming requests and decides whether the system should make a prediction, run an evaluation, compare results, explain a prediction, retrain the model, or check system status.

The DataPipeline is responsible for getting the raw coin data ready. It downloads the data, checks that it is valid, builds useful features, and splits everything into training and testing sets. After that, the RobustScaler comes in to normalize the data so the model receives inputs in a consistent format, which helps improve performance.

The ModelEngine is where the main prediction work happens. It loads saved models, runs predictions, evaluates how well the model performs, and applies any thresholds or rounding needed before returning results. The outputs are packaged into PredictResponse objects, which include things like the predicted direction, confidence level, probabilities, and expected price changes. These also include detailed forecast points with timestamps and value ranges.

The IncrementalTrainer keeps the system up to date over time. It checks whether retraining is needed, manages how the data is split based on time, and saves updated models and metadata so the system can keep learning as new data comes in. For explainability, the ShapEngine calculates feature importance values so it is easier to understand why the model made a certain prediction. It also caches these results to make future explanations faster.

Finally, evaluation results are returned as structured responses with metrics such as error rates and accuracy scores. Altogether, the system is designed to be modular and flexible, combining data processing, prediction, evaluation, retraining, and interpretability into one cohesive workflow.

4.6 GUI Design

The GUI design for the Hybrid CNN-LSTM + BiLSTM Cryptocurrency Prediction System was approached with a clear priority: the interface should get out of the way and let users focus on the predictions. [14] That means a clean layout, intuitive navigation, and visual outputs that communicate market trends at a glance without requiring the user to understand what is happening inside the model. [20] The interface is built using React, which provides the flexibility and component reusability needed to build a responsive, API-connected financial dashboard.

The key design decisions include:

- A dark-themed, SaaS-style financial dashboard that reduces visual fatigue during extended analysis sessions and gives the interface a professional character.

- Glassmorphism-based UI components that create a layered, visually clean aesthetic without feeling overly decorative.
- Gradient accent colors running from purple to cyan, used consistently to highlight key actions, prediction outputs, and interactive elements.
- Smooth animations and transitions powered by Framer Motion, which makes the interface feel responsive without being distracting.
- A centralized navigation structure that provides quick access to the landing page, dashboard, and model information sections from anywhere in the app.
- An interactive coin selection dropdown covering BTC, ETH, BNB, XRP, ADA, SOL, and DOGE, allowing users to switch assets without reloading the page.
- A clearly prominent “Predict” button designed as the primary call-to-action, triggering the forecasting pipeline with a single click.
- Multiple output panels arranged in a logical dashboard layout, including:
 1. A live price prediction chart rendered with Recharts.
 2. A directional prediction indicator (STRONG UP / NEUTRAL / STRONG DOWN).
 3. A confidence score visualization showing how certain the model is.
 4. A sentiment panel displaying the current Fear and Greed Index value.
 5. A model performance panel showing evaluation metrics.
- Modular, reusable UI components built for consistency and easy maintenance:
 1. GlassCard (container component for consistent UI layout)
 2. PredictionCard (displays model output results)
 3. MetricsPanel (shows MAPE, RMSE, F1, and other evaluation metrics)
 4. SentimentPanel (visualizes the market sentiment index)
 5. ForecastChart (plots actual versus predicted price trends)
- A fully responsive layout that works across desktop, tablet, and mobile screen sizes.
- API caching to keep load times short and prediction responses snappy.
- A minimal-click interaction flow: users can generate a prediction in no more than two actions.

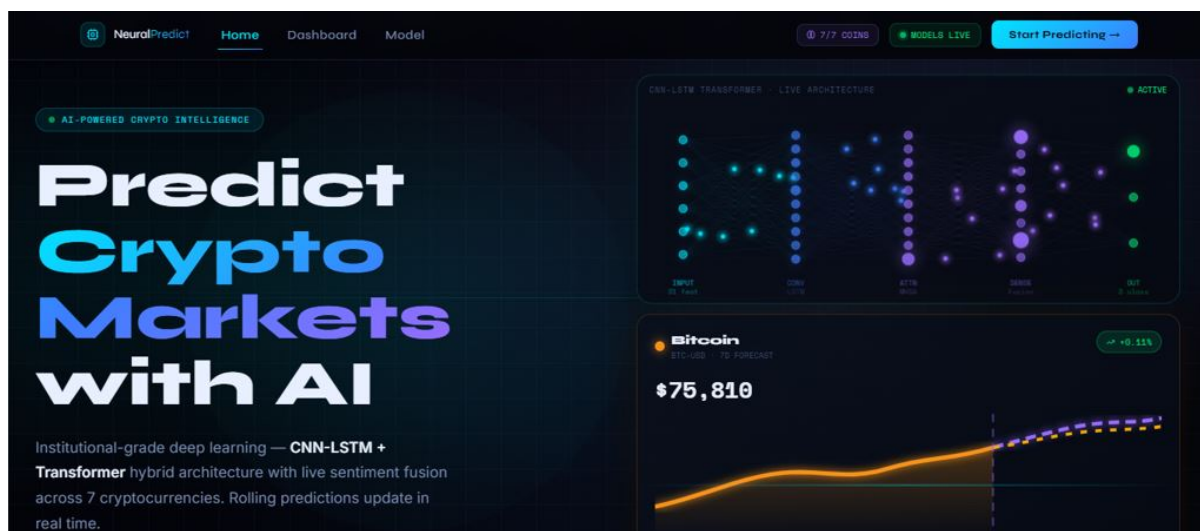


Figure 4.9: Main Home Page of the cryptocurrency prediction interface

Figure 4.9 shows the main landing page of NeuralPredict, the AI-powered cryptocurrency intelligence platform. The dark-themed interface features a top navigation bar with Home, Dashboard, and Model sections, alongside live model status and prediction credit indicators. The central area leads with the headline “Predict Crypto Markets with AI,” positioning the platform’s core value immediately. Below that, a brief description explains the institutional-grade hybrid architecture combining CNN, LSTM, and BiLSTM models with real-time sentiment fusion. A neural network architecture diagram on the right illustrates data flow from input through convolutional LSTM and attention layers to the dense output. A live Bitcoin forecast preview at the bottom shows current pricing alongside projected trends. The “Start Predicting” button provides the primary call-to-action, and users can monitor model activity and performance in real time directly from the landing page.



Figure 4.10: Main Dashboard of Cryptocurrency Prediction

Figure 4.10 shows the Prediction Dashboard of NeuralPredict. The dark interface retains the navigation bar from the landing page and adds a live ticker displaying real-time prices for Bitcoin, Ethereum, XRP, and other major assets. The main panel focuses on Bitcoin (BTC/USD), showing the current price alongside a 7-day forecast visualization. Users can switch currencies from a dropdown and select prediction horizons (3D, 5D, 7D, 14D, 30D) directly from the interface. The central chart overlays actual price movements with both CNN-LSTM and BiLSTM model predictions and forecasted trend lines. A right-side panel displays performance metrics including MAPE, directional accuracy, and F1 score, along with SHAP analysis access for model interpretability. The “Run Prediction” button and a retraining option sit prominently within the panel, while live model indicators reflect real-time inference status.

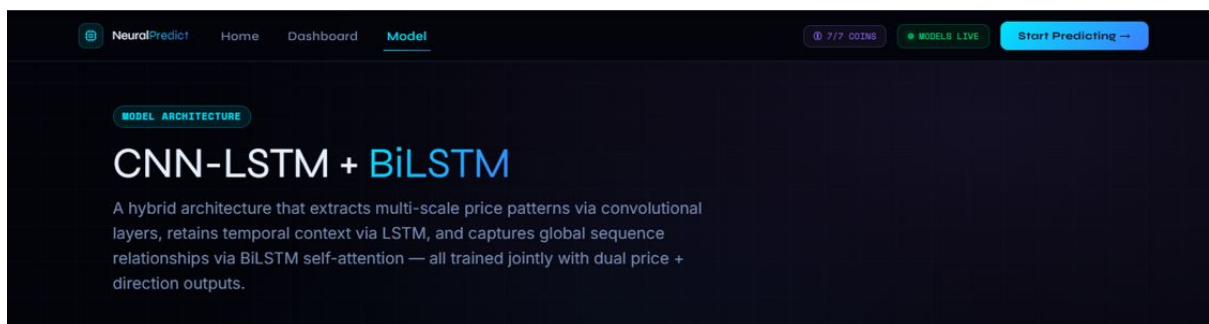


Figure 4.11: CNN-LSTM + BiLSTM Architecture Description



Figure 4.12: CNN-LSTM + BiLSTM Hybrid Model Architecture Diagram

Figures 4.11 and 4.12 present the hybrid deep learning model that powers NeuralPredict. The model takes an input of shape (30×31) , representing 30 time steps with 31 features each, and routes it through two parallel branches designed to capture different aspects of the data. The first branch is a CNN-LSTM pipeline: convolution and pooling layers extract local temporal patterns, and an LSTM layer then models short-term sequential dependencies from those extracted features. The second branch uses a stacked BiLSTM with self-attention to capture longer-range relationships and identify which time steps are most relevant to the current prediction.

Outputs from both branches are merged into a shared dense representation (Dense(128) to Dense(64)), which feeds into two separate output heads. One head performs price regression using Huber loss for robustness against outliers. The other predicts mar-

ket direction (uptrend, downtrend, neutral) using a softmax classifier trained with focal loss ($\gamma = 2$) to handle the natural class imbalance in directional labels. The architecture is multi-branch and multi-task by design, combining convolutional, recurrent, and attention-based methods to improve both price forecasting accuracy and directional prediction across volatile cryptocurrency markets.

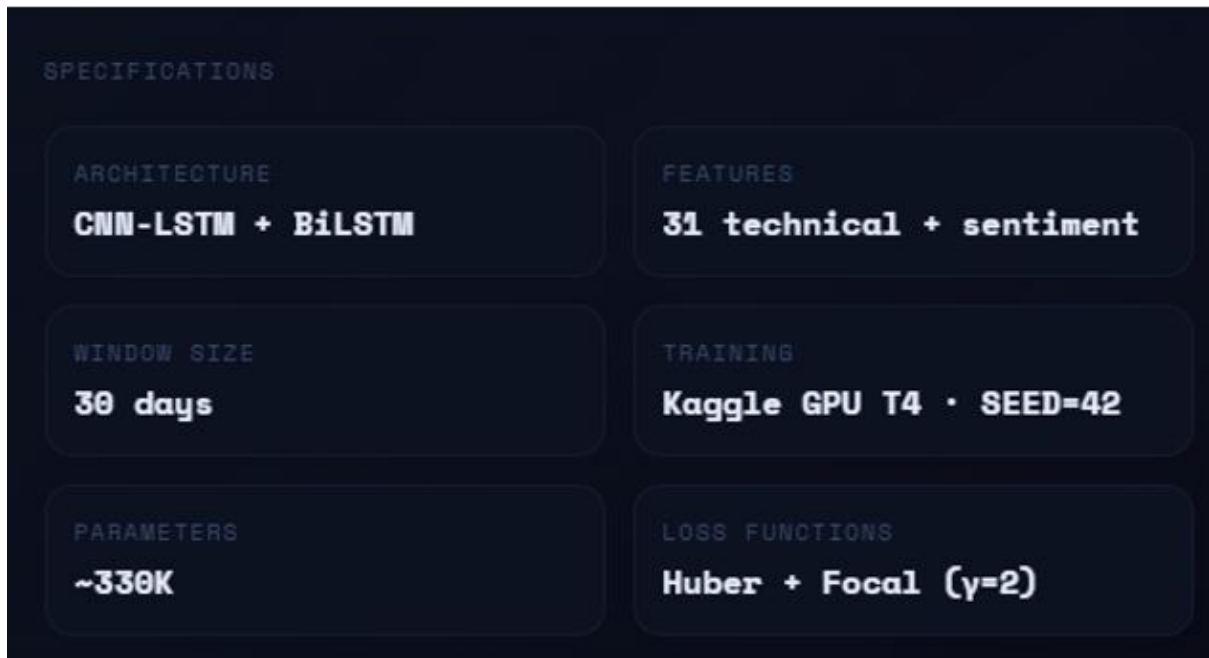


Figure 4.13: Model Specifications of Cryptocurrency Prediction

Figure 4.13 summarizes the key specifications of the NeuralPredict model. The hybrid CNN-LSTM and BiLSTM framework is designed to capture both local and long-range temporal dependencies in cryptocurrency time-series data. The model works on a 30-day input window using 31 combined technical and sentiment-derived features. Training is conducted on Kaggle’s GPU (T4) environment with a fixed random seed (42) for reproducibility. The architecture has approximately 350K trainable parameters, balancing model capacity with training efficiency. Loss functions include Huber loss for price regression and focal loss ($\gamma = 2$) for direction classification, a combination chosen to handle both price-scale outliers and directional class imbalance in a principled way.

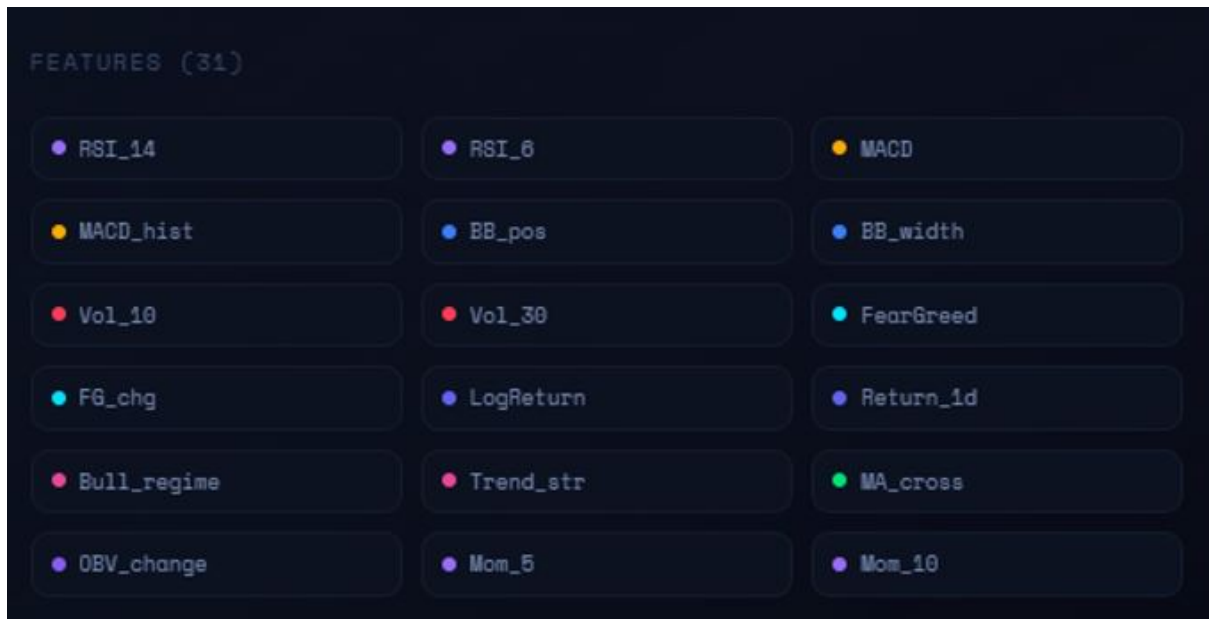


Figure 4.14: Feature Set Overview of the Cryptocurrency Prediction

Figure 4.14 shows the 18 core input features used by NeuralPredict, spanning several analytical dimensions to give the model a comprehensive view of market conditions. Momentum is represented through RSI_14, RSI_6, MACD, and MACD_hist. Volatility is captured via BB_pos and BB_width. Volume dynamics come in through Vol_10, Vol_30, and OBV_change. Return-based signals include LogReturn and Return_1d. Market sentiment enters through FearGreed and FG_chg, while broader structural context is provided by Bull_regime, Trend_str, and MA_cross. Short-to-medium term momentum is further represented by Mom_5 and Mom_10.

Benchmark Results					
Kaggle GPU T4 · SEED=42 · Test split 15%					
COIN	DA STRONG (%)	DA CONFIDENT (%)	MAPE (%)	F1 MACRO	RMSE (\$)
BTC -USD	61.7	70.8	1.68	0.5839	1,842
ETH -USD	67.0	50.0	2.79	0.4010	180
BNB -USD	63.2	56.5	1.97	0.3870	22
XRP -USD	50.9	47.6	2.92	0.4910	0.0400
ADA -USD	61.7	60.4	3.48	0.5980	0.0300
SOL -USD	72.1	58.8	3.10	0.4100	11
DOGE -USD	57.4	48.4	3.00	0.5510	0.0100

Figure 4.15: Benchmark Results Across Seven Cryptocurrency Pairs

Figure 4.15 presents benchmark evaluation results across all seven cryptocurrency pairs, tested on a 15 percent holdout set using a Kaggle T4 GPU with a fixed seed of 42. Five metrics are reported: Directional Accuracy Strong (DA Strong), Directional

Accuracy Confident (DA Confident), MAPE, F1 Macro, and RMSE.

SOL-USD achieves the highest DA Strong at 72.1 percent, while BTC-USD leads on DA Confident at 70.8 percent, reflecting stronger directional predictability for high-liquidity assets with more institutional trading activity. ETH-USD follows with a DA Strong of 67.0 percent, though its DA Confident drops to 50.0 percent, suggesting more uncertainty in high-confidence predictions specifically. ADA-USD records the best F1 Macro score at 0.5980, indicating the most balanced classification performance across directional classes. XRP-USD and DOGE-USD show weaker results, with DA Strong values of 50.9 percent and 57.4 percent respectively, reflecting the harder-to-predict nature of more sentiment-driven assets. RMSE values vary substantially across coins, largely driven by differences in absolute price scale, with BTC-USD naturally recording the highest at \$1,842.

4.6.1 Bitcoin (BTC-USD) Price Prediction

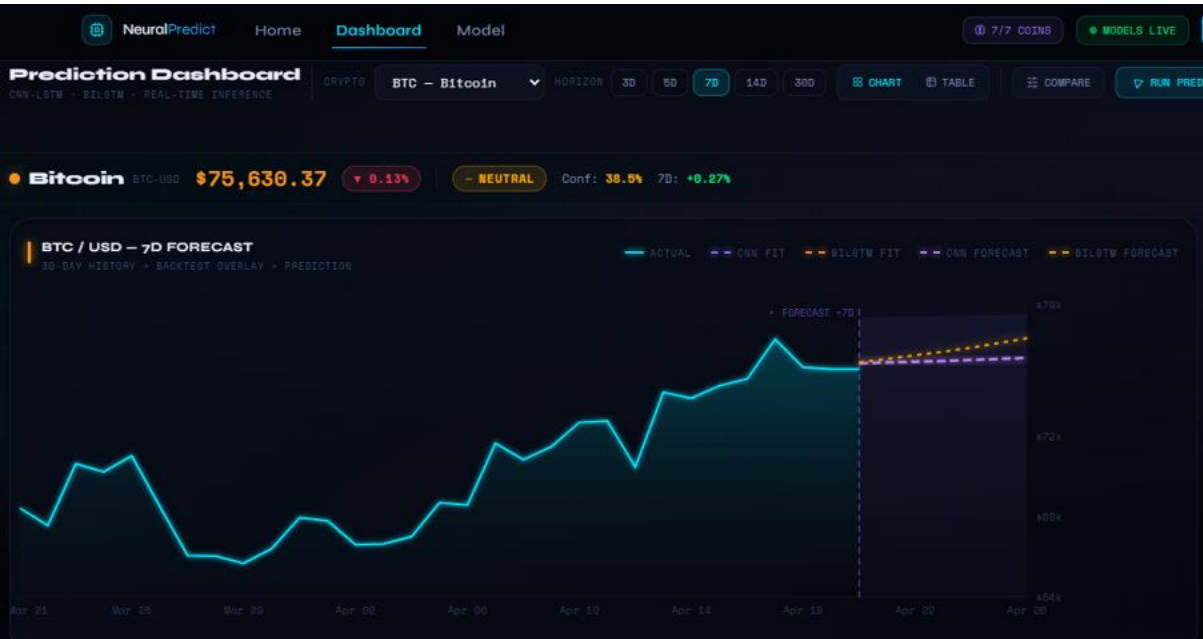


Figure 4.16: BTC/USD 7-Day Forecast Chart

BTC FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr 20	\$75,922.23	\$75,981.57	38.5%	HOLD
Apr 21	\$75,969.47	\$76,143.47	37.0%	HOLD
Apr 22	\$76,003.64	\$76,326.05	35.5%	HOLD
Apr 23	\$76,042.74	\$76,501.25	34.0%	HOLD
Apr 24	\$76,093.91	\$76,700.4	32.5%	HOLD
Apr 25	\$76,141.86	\$76,938.68	31.0%	HOLD
Apr 26	\$76,192.51	\$77,174.17	29.5%	HOLD

Figure 4.17: BTC/USD 7-Day Tabular Forecast Output

Figures 4.16 and 4.17 show the dual-view forecasting output for Bitcoin, combining a visual chart overlay with a structured daily prediction table. Both the CNN-LSTM and BiLSTM models project a gradual upward trajectory for Bitcoin over the 7-day horizon, with reasonable agreement between the two, particularly in the near-term days. The declining confidence trend across the forecast window reflects the model’s built-in awareness that uncertainty grows the further out you project. The consistent HOLD signal across all forecast days suggests a cautious, range-bound market outlook at the time of inference. This dual-output design improves interpretability by giving users not just predicted price levels but also a sense of how confident and directionally consistent those predictions are.

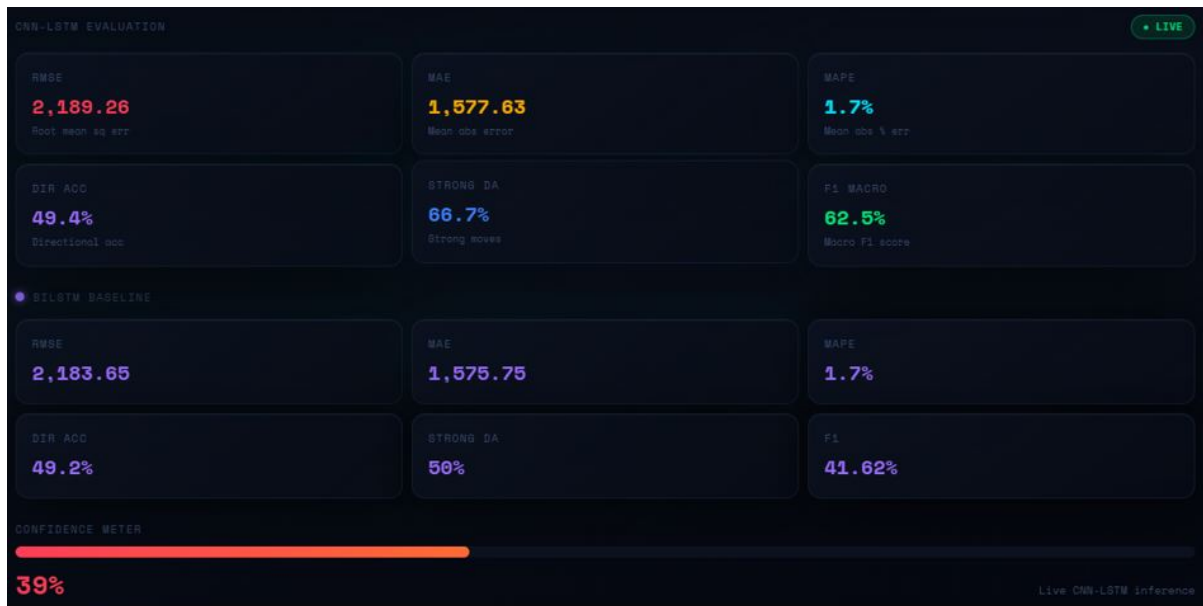


Figure 4.18: Live Model Evaluation Metrics for BTC-USD — CNN-LSTM and BiLSTM

Figure 4.18 presents live evaluation metrics for both the CNN-LSTM and BiLSTM baseline models on BTC-USD prediction, alongside a confidence meter reflecting the current inference state. The CNN-LSTM records an RMSE of \$2,189.26 and MAE of \$1,577.63, with a MAPE of 1.7 percent, indicating low percentage error relative to Bitcoin’s price scale. Directional accuracy is 49.4 percent overall, but Strong DA reaches 66.7 percent and the Macro F1 score is 62.5 percent, suggesting reasonably balanced classification performance on high-conviction market moves.

The BiLSTM Baseline produces similar regression metrics: RMSE of \$2,183.65 and MAE of \$1,575.75, again at 1.7 percent MAPE. However, its directional accuracy of 49.2 percent and Strong DA of 50.0 percent are notably weaker, with an F1 score of 41.62 percent, pointing to more difficulty with directional classification despite comparable price-level accuracy.

The Confidence Meter registers 39 percent, derived from the live CNN-LSTM inference. Overall, both models achieve similar regression performance, but the CNN-LSTM holds a clear advantage in directional classification, particularly on strong market moves, which is exactly the output most relevant for generating actionable trading signals.



Figure 4.19: SHAP Explainability for CNN-LSTM Model (BTC-USD)



Figure 4.20: SHAP Explainability for BiLSTM Model (BTC-USD)

Figures 4.19 and 4.20 reveal meaningful differences in how the CNN-LSTM and BiLSTM models interpret the feature space when predicting Bitcoin price movements. The CNN-LSTM distributes importance relatively evenly across volume, volatility, and momentum features, while the BiLSTM places disproportionately high weight on Vol_regime and FearGreed, suggesting it is more sensitive to broader sentiment conditions and structural regime shifts. This contrast makes sense given the architectural differences: the CNN-LSTM’s convolutional layers naturally pick up localized price patterns, while the BiLSTM’s bidirectional structure appears more receptive to longer-range sentiment and regime dynamics. Understanding these differences goes beyond academic interest. It helps build confidence in the system’s predictions and clarifies which model is best suited

to which type of market environment.

4.6.2 Ethereum (ETH-USD) Price Prediction



Figure 4.21: ETH/USD 7-Day Visual Forecast Output

ETH FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr 20	\$2,309.01	\$2,312.02	38.2%	HOLD
Apr 21	\$2,309.42	\$2,317.88	36.7%	HOLD
Apr 22	\$2,310.05	\$2,325.81	35.0%	HOLD
Apr 23	\$2,310.01	\$2,333.46	33.7%	HOLD
Apr 24	\$2,311.19	\$2,335.13	32.2%	HOLD
Apr 25	\$2,311.81	\$2,335.99	30.7%	HOLD
Apr 26	\$2,312.46	\$2,336.64	29.2%	HOLD

Figure 4.22: ETH/USD 7-Day Tabular Forecast Output

Figures 4.21 and 4.22 demonstrate the integrated dual-output design of NeuralPredict, where graphical and tabular forecasts work together to provide a clearer picture of market expectations. Both models suggest relatively stable price movement for Ethereum over the 7-day horizon, with only modest upward inclination and strong agreement between their outputs. The decreasing confidence values across the forecast period reflect growing uncertainty as projections extend further out, while the consistent HOLD signals point to a conservative market read at the time of inference. This combined presentation supports more informed decision-making by giving users both an intuitive visual trend and the underlying predictive metrics in one view.



Figure 4.23: Live Model Evaluation Metrics for ETH-USD — CNN-LSTM and BiLSTM

Figure 4.23 compares the CNN-LSTM model against the BiLSTM baseline on Ethereum prediction. The CNN-LSTM achieves slightly lower error values, with RMSE of 117.05, MAE of 79.76, and MAPE of 2.76 percent, reflecting marginally better price-level accuracy. Directional accuracy for both models sits close to 50 percent, indicating limited ability to reliably call the direction of movement. However, the CNN-LSTM performs better on strong directional moves at 65.1 percent versus 62.3 percent. Despite this, its F1 macro score of 39.43 percent comes in lower than the baseline, pointing to weaker overall classification balance. The confidence meter at 38 percent signals limited prediction reliability in this context. The CNN-LSTM offers modest improvements in regression but faces real challenges in directional and classification performance on ETH.



Figure 4.24: SHAP Explainability for CNN-LSTM Model (ETH-USD)



Figure 4.25: SHAP Explainability for BiLSTM Model (ETH-USD)

Figures 4.24 and 4.25 compare SHAP global feature importance for CNN-LSTM and BiLSTM models on Ethereum prediction. The CNN-LSTM is primarily shaped by volatility and technical indicators such as BB_width, Vol_change, and SMA-based trends, reflecting a volatility-sensitive and indicator-driven approach. The BiLSTM is dominated by return-based features like Return_14d and LogReturn, pointing to a stronger reliance on historical price momentum. Both models treat volatility as a risk signal, but CNN-LSTM places greater emphasis on structural market condition indicators while BiLSTM prioritizes return patterns. Bullish drivers in CNN-LSTM are trend-oriented; in BiLSTM they are momentum-driven. The figures together highlight a fundamental difference in learning behavior: CNN-LSTM captures structural and volatility signals while BiLSTM

focuses on temporal return dependencies, showing how complementary the two architectures actually are.

4.6.3 Binance Coin (BNB-USD) Price Prediction



Figure 4.26: BNB/USD 7-Day Visual Forecast Output

BNB FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr. 20	\$624.194	\$623.367	38.3%	HOLD
Apr. 21	\$624.497	\$622.453	36.6%	HOLD
Apr. 22	\$624.776	\$621.411	35.3%	HOLD
Apr. 23	\$625.032	\$620.242	33.6%	HOLD
Apr. 24	\$625.284	\$619.211	32.3%	HOLD
Apr. 25	\$625.557	\$618.234	30.6%	HOLD
Apr. 26	\$625.829	\$617.483	29.3%	HOLD

Figure 4.27: BNB/USD 7-Day Tabular Forecast Output

Figures 4.26 and 4.27 show the prediction dashboard for Binance Coin, combining historical price visualization with a 7-day forecast generated by both CNN-LSTM and BiLSTM models. The chart displays recent price movements alongside projected trends, while the table below provides a day-by-day breakdown of predicted values, confidence levels, and trading signals. Overall, the system points to a relatively stable outlook with moderate uncertainty and consistent HOLD recommendations across the forecast horizon.



Figure 4.28: Live Model Evaluation Metrics for BNB-USD — CNN-LSTM and BiLSTM

Figure 4.28 presents evaluation results for CNN-LSTM and BiLSTM models on Binance Coin. The CNN-LSTM shows slightly improved accuracy with lower RMSE and MAE values, while both models produce similar MAPE and directional accuracy figures, reflecting comparable ability to track overall price trends. CNN-LSTM achieves higher strong directional accuracy and F1 score, demonstrating an edge in identifying significant market movements. The confidence meter at approximately 38 percent indicates moderate reliability, pointing to some residual uncertainty in the predictions. Overall, the CNN-LSTM delivers better predictive performance and robustness than the BiLSTM baseline for BNB, even if neither model provides highly decisive signals.



Figure 4.29: SHAP Explainability for CNN-LSTM Model (BNB-USD)



Figure 4.30: SHAP Explainability for BiLSTM Model (BNB-USD)

Figures 4.29 and 4.30 present a consolidated view of SHAP global feature importance from both model variants for BNB-USD. Across both, volatility measures (Vol_10, Vol_30, Vol_ratio) and price range (HL_range) emerge as dominant risk drivers. Short-term return features (Return_1d, Return_3d, Return_7d) consistently act as strong bullish signals, capturing momentum effects. Moving averages (Close_SMA10, Close_SMA30, SMA10_SMA30) also play a major role, reflecting trend-following behavior baked into both models. Sentiment through FearGreed is moderately influential in both cases. MACD and its histogram and signal components contribute to both bullish and risk interpretations depending on context. The second model places greater emphasis on longer-term trend signals (Close_SMA30) and volume dynamics (OBV_change). Across both figures, the models show a consistent reliance on a combination of momentum, volatility, and trend indicators to drive their predictions.

4.6.4 XRP-USD Price Prediction



Figure 4.31: XRP/USD 7-Day Visual Forecast Output

XRP FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr. 20	\$1.4232	\$1.4184	40.5%	HOLD
Apr. 21	\$1.4268	\$1.4185	39.0%	HOLD
Apr. 22	\$1.4307	\$1.4153	37.5%	HOLD
Apr. 23	\$1.4358	\$1.4151	36.0%	HOLD
Apr. 24	\$1.4423	\$1.4168	34.5%	HOLD
Apr. 25	\$1.4494	\$1.4193	33.0%	HOLD
Apr. 26	\$1.4574	\$1.4228	31.5%	HOLD

Figure 4.32: XRP/USD 7-Day Tabular Forecast Output

Figures 4.31 and 4.32 show the XRP/USD 7-day forecast from NeuralPredict. Historical price trends are displayed alongside projections from both models, with the forecast region clearly distinguished from past data. Both CNN-LSTM and BiLSTM indicate a gradual upward trend, though slight differences exist in the predicted values and growth rates. The current price sits around \$1.41, with neutral market sentiment and a confidence level near 40 percent. CNN-LSTM outputs are marginally higher than BiLSTM across the forecast period. Confidence decreases steadily day by day, reflecting the natural growth of uncertainty as projections extend further. All days carry a HOLD signal, indicating no strong buy or sell recommendation. Overall, the forecast suggests a modest

upward outlook with cautious interpretation warranted given the relatively low confidence levels.

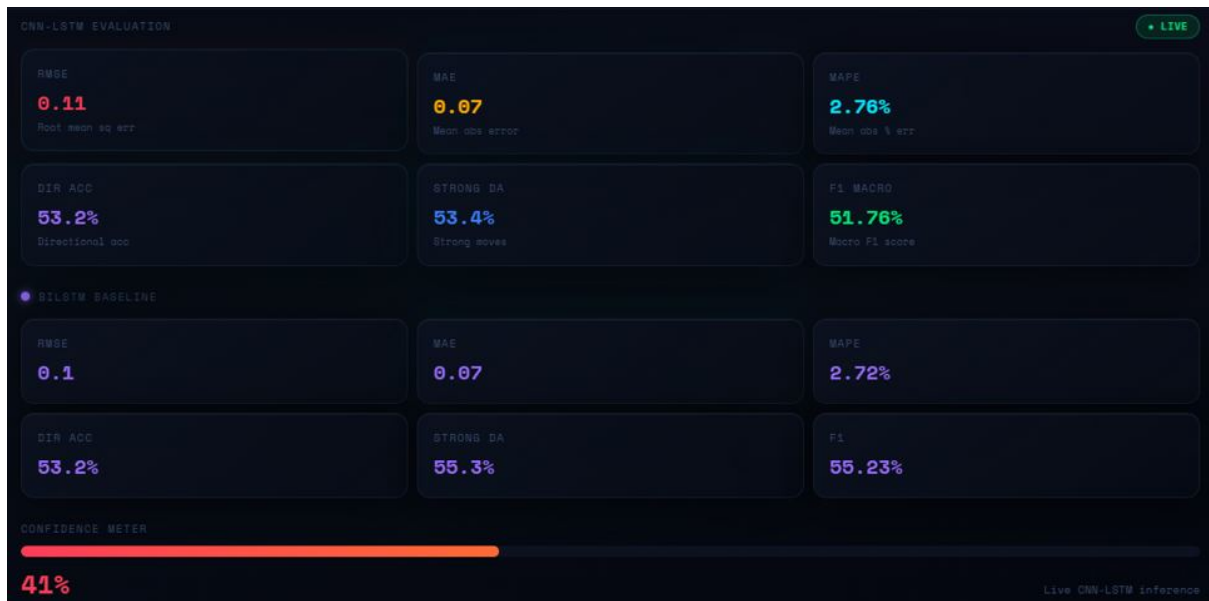


Figure 4.33: Live Model Evaluation Metrics for XRP-USD — CNN-LSTM and BiLSTM

Figure 4.33 presents the performance evaluation for XRP-USD. The CNN-LSTM records an RMSE of 0.11, MAE of 0.07, and MAPE of 2.76 percent, reflecting relatively low price-level error. Directional accuracy stands at 53.2 percent with a Strong DA of 53.4 percent and an F1 macro score of 51.76 percent, indicating moderate classification performance. The BiLSTM baseline achieves a slightly lower RMSE of 0.10, the same MAE of 0.07, and a marginally better MAPE of 2.72 percent. Both models share the same overall directional accuracy, but BiLSTM edges ahead in Strong DA at 55.3 percent and F1 score at 55.23 percent. The confidence meter shows 41 percent, suggesting moderate reliability. For XRP, the results are closely matched between models, with BiLSTM holding a slight advantage in classification-related metrics.



Figure 4.34: SHAP Explainability for CNN-LSTM Model (XRP-USD)



Figure 4.35: SHAP Explainability for BiLSTM Model (XRP-USD)

Figures 4.34 and 4.35 provide a unified view of SHAP global feature importance from both live models on XRP-USD. In both cases, Close_SMA10 stands out as the most influential feature, underscoring the importance of short-term trend direction. This is followed by Return_3d and Return_14d, showing that both recent and medium-term price movements carry significant predictive weight. The FearGreed index and Mom_10 highlight the additional impact of sentiment and momentum. MACD_hist and Bull_regime contribute moderately, indicating that broader market conditions also factor in. Lower-ranked features including Return_7d, MA_cross, Price_pos, BB_pos, and Vol_30 play a supporting role in refining the final outputs. Bullish drivers are dominated by trend and

return-based signals, while risk drivers emphasize volatility and position-related indicators. Both models tell a consistent story: trend and momentum signals drive the most important predictions, with market risk factors providing secondary context.

4.6.5 Cardano (ADA-USD) Price Prediction



Figure 4.36: ADA/USD 7-Day Visual Forecast Output

ADA FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr 20	\$0.243079	\$0.244921	44.4%	HOLD
Apr 21	\$0.240189	\$0.244346	42.9%	HOLD
Apr 22	\$0.238536	\$0.243811	41.4%	HOLD
Apr 23	\$0.236025	\$0.243876	39.9%	HOLD
Apr 24	\$0.233399	\$0.244079	38.4%	HOLD
Apr 25	\$0.230877	\$0.244050	36.9%	HOLD
Apr 26	\$0.228438	\$0.243963	35.4%	HOLD

Figure 4.37: ADA/USD 7-Day Tabular Forecast Output

Figures 4.36 and 4.37 together present the prediction dashboard for Cardano (ADA/USD). The interface shows the current price, a short-term percentage change, and an overall neutral signal with moderate confidence at the top. The main chart visualizes the 7-day forecast, combining historical price movement with projected trends from both models. The actual price appears in bright cyan, while the forecast paths from CNN-LSTM and BiLSTM are shown as dotted lines extending into the forecast period. A vertical marker clearly separates past data from predictions, making the transition easy to read. The projection suggests a slight downward lean over the coming days. Below the chart, the table lists daily forecasts from both models alongside confidence scores that gradually

decline, indicating growing uncertainty. The signal column shows consistent HOLD recommendations throughout, reflecting a cautious market outlook.

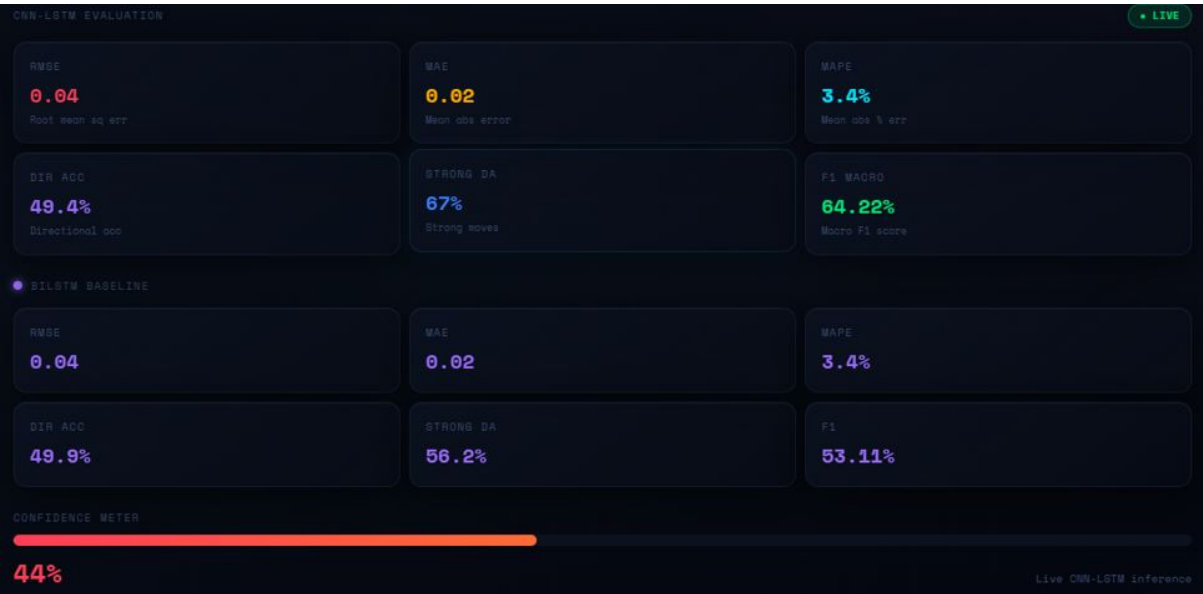


Figure 4.38: Live Model Evaluation Metrics for ADA-USD — CNN-LSTM and BiLSTM

Figure 4.38 shows the performance evaluation for ADA-USD. The CNN-LSTM produces low RMSE and MAE values alongside a MAPE of 3.4 percent, indicating solid overall accuracy relative to Cardano’s price scale. Directional accuracy sits at around 49.4 percent, nearly balanced between up and down calls, but Strong DA reaches 67 percent, suggesting the model performs considerably better when market moves are pronounced. The F1 macro score of 64.22 percent reflects reasonable balance between precision and recall across prediction classes. The BiLSTM baseline achieves similar error metrics but falls slightly short on both Strong DA and F1 score, indicating that the CNN-LSTM offers better classification strength when it matters most. The confidence meter at 44 percent reflects a cautious but not dismissive level of certainty in the predictions.



Figure 4.39: SHAP Explainability for CNN-LSTM Model (ADA-USD)



Figure 4.40: SHAP Explainability for BiLSTM Model (ADA-USD)

Figures 4.39 and 4.40 compare SHAP global feature importance from live analysis across both model architectures for ADA-USD. In the CNN-LSTM view, momentum and trend indicators dominate: SMA10-SMA30 crossovers and log returns are the most influential features, reflecting a strong reliance on short-term directional signals. Crossover features and volatility metrics including MACD and volume contribute notably as risk-related factors. The BiLSTM view tells a different story, placing more emphasis on MACD signals, Bollinger Band width, and price range, reflecting greater sensitivity to volatility and market expansion dynamics. Log returns remain important across both models, reinforcing their role as a consistent signal for capturing price momentum. Bullish drivers are mainly trend and momentum-based in both cases, while risk drivers stem from volatility,

volume shifts, and regime changes. The comparison captures the difference well: CNN-LSTM is more trend-focused while BiLSTM takes a broader view of market dynamics.

4.6.6 Solana (SOL-USD) Price Prediction

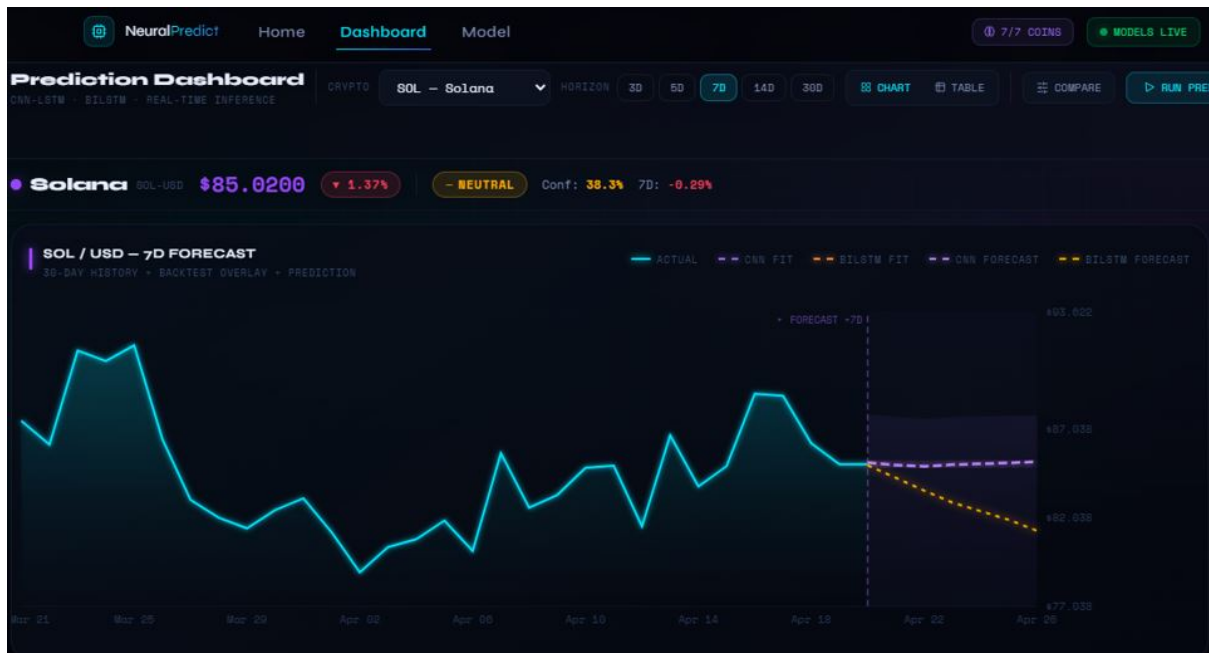


Figure 4.41: SOL/USD 7-Day Visual Forecast Output

SOL FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr 20	\$85.1145	\$84.9610	38.3%	HOLD
Apr 21	\$84.8719	\$84.2598	36.5%	HOLD
Apr 22	\$84.9072	\$83.5430	35.3%	HOLD
Apr 23	\$85.0028	\$82.8733	33.8%	HOLD
Apr 24	\$85.0517	\$82.3853	32.3%	HOLD
Apr 25	\$85.0973	\$81.9032	30.8%	HOLD
Apr 26	\$85.1636	\$81.2859	29.3%	HOLD

Legend: CNN-LSTM (blue), BiLSTM (yellow)

Footer: 7 DAYS | REAL MODEL OUTPUT

Figure 4.42: SOL/USD 7-Day Tabular Forecast Output

Figures 4.41 and 4.42 present the prediction dashboard for Solana, combining live pricing with short-term forecast analytics. The current price is shown at around \$85.02, with a slight negative change and an overall neutral sentiment at moderate confidence. The 7-day forecast graph shows recent price fluctuations stabilizing near the present level, with CNN-LSTM and BiLSTM projections both suggesting a mild downward lean over the coming days. Shaded uncertainty regions around the forecasts reflect how prediction confidence widens as the horizon extends. The tabular breakdown below covers daily predictions from April 20 to April 26, with confidence levels declining progressively. All days carry a HOLD signal, indicating no strong directional trading recommendation.

Together, the figures give a clear and concise view of short-term market expectations for SOL alongside the model’s level of certainty in those projections.



Figure 4.43: Live Model Evaluation Metrics for SOL-USD — CNN-LSTM and BiLSTM

Figure 4.43 summarizes the evaluation results for SOL-USD. The CNN-LSTM shows slightly higher RMSE and MAE compared to the BiLSTM baseline, but the differences are marginal and both models maintain similar MAPE figures close to 3 percent. Where CNN-LSTM clearly stands out is in strong directional accuracy and F1 macro score, indicating better performance when identifying significant market moves. The BiLSTM baseline shows weaker signal quality on these classification-oriented metrics despite comparable overall accuracy. The confidence meter at 38 percent reflects moderate certainty, with room for improvement in prediction robustness. The overall picture is that CNN-LSTM offers better signal quality for Solana, even if neither model achieves highly decisive directional predictions.



Figure 4.44: SHAP Explainability for CNN-LSTM Model (SOL-USD)



Figure 4.45: SHAP Explainability for BiLSTM Model (SOL-USD)

Figures 4.44 and 4.45 compare SHAP feature importance for both models on SOL-USD. For the CNN-LSTM in Figure 4.43, the model draws on a diverse mix of technical signals: RSI_14 and short-term moving averages (Close_SMA10, Close_SMA50) are the strongest contributors, with volume signals (Vol_change, Vol_regime, Vol_30) and short-horizon returns also playing a meaningful role. The bullish drivers panel highlights momentum and trend-following features, while risk drivers emphasize volatility and volume spikes. The BiLSTM in Figure 4.44 shows a very different and far more concentrated importance structure: Mom_10 and Price_pos together account for nearly all of the model's explanatory power, with most other features contributing negligibly. This points to a simplified, momentum-focused decision process rather than the broader multi-factor view the

CNN-LSTM develops. Side by side, the figures illustrate a real architectural difference: CNN-LSTM captures a richer signal landscape across multiple feature categories, while the BiLSTM for Solana behaves more like a focused momentum model.

4.6.7 Dogecoin (DOGE-USD) Price Prediction

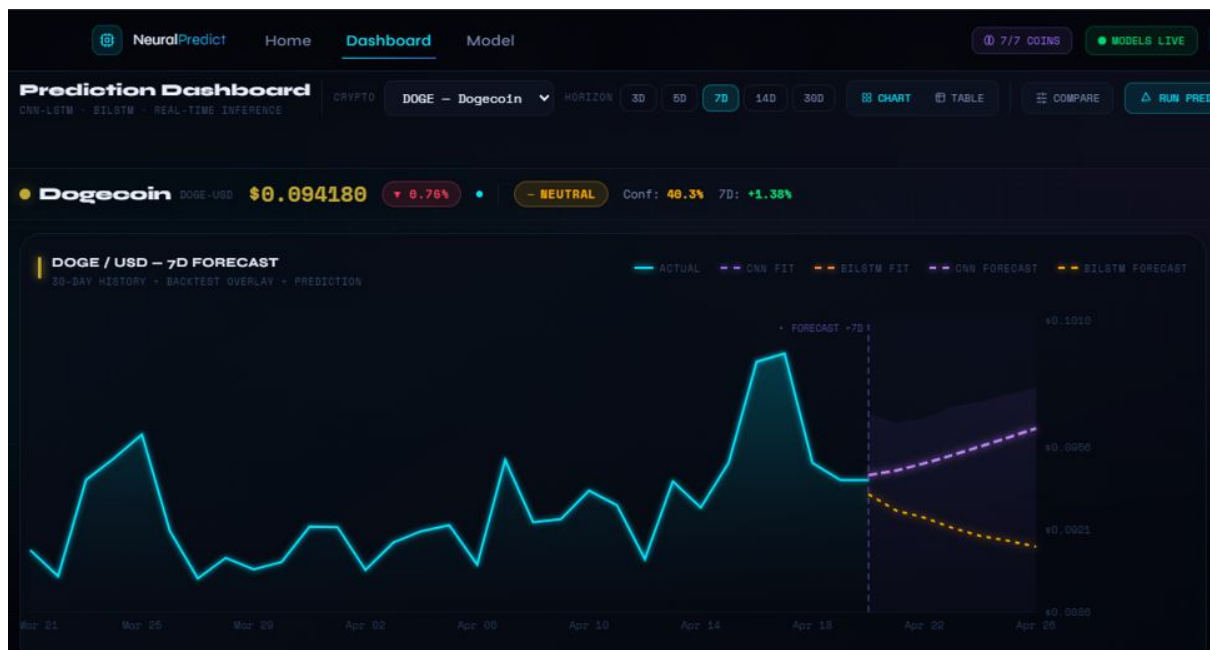


Figure 4.46: DOGE/USD 7-Day Visual Forecast Output

DOGE FORECAST - 7-DAY HORIZON				
DATE	CNN-LSTM	BiLSTM	CONFIDENCE	SIGNAL
Apr 20	\$0.094389	\$0.093578	40.3%	HOLD
Apr 21	\$0.094586	\$0.092888	38.8%	HOLD
Apr 22	\$0.094889	\$0.092589	37.3%	HOLD
Apr 23	\$0.095234	\$0.092157	35.6%	HOLD
Apr 24	\$0.095604	\$0.091815	34.3%	HOLD
Apr 25	\$0.095982	\$0.091610	32.6%	HOLD
Apr 26	\$0.096374	\$0.091360	31.3%	HOLD

Figure 4.47: DOGE/USD 7-Day Tabular Forecast Output

Figures 4.46 and 4.47 present the prediction dashboard for Dogecoin over a 7-day horizon. The chart displays recent price movements alongside model-generated projections, with a clear transition from historical data to forecast territory. The two models diverge here in a notable way: the CNN-LSTM projection suggests a mild upward trend while the BiLSTM indicates a gradual decline, which itself is worth paying attention to as a sign of model uncertainty on this asset. The current price is around \$0.094, with a neutral signal and a confidence level near 40 percent. A shaded confidence region around the forecasts highlights how much uncertainty exists in either direction. The daily forecast table below confirms this divergence, with CNN-LSTM outputs consistently above BiLSTM estimates

across the forecast period. Confidence decreases steadily over time, and all days are assigned a HOLD signal, reflecting cautious market conditions. Together, the visuals and table give a balanced view of short-term DOGE expectations with appropriate caveats around the level of certainty.



Figure 4.48: Live Model Evaluation Metrics for DOGE-USD — CNN-LSTM and BiLSTM

Figure 4.48 shows the performance evaluation for DOGE-USD. Both models achieve low RMSE and MAE values around 0.01, meaning their predictions stay close to actual price movements in absolute terms. MAPE values sit between 3.4 and 3.5 percent for both, reflecting reasonable percentage accuracy given Dogecoin’s price level. Directional accuracy hovers near 50 percent for both models, confirming that correctly calling the direction of movement remains genuinely uncertain for this asset. CNN-LSTM holds a slight edge in strong directional accuracy and F1 macro score, suggesting a marginal advantage in identifying more decisive price shifts. The confidence meter at around 40 percent reinforces the cautious picture. Both models perform similarly overall, with CNN-LSTM offering subtle improvements but neither achieving highly confident or consistent directional signals on Dogecoin.



Figure 4.49: SHAP Explainability for CNN-LSTM Model (DOGE-USD)



Figure 4.50: SHAP Explainability for BiLSTM Model (DOGE-USD)

Figures 4.49 and 4.50 present SHAP global feature importance from both models on DOGE-USD. The CNN-LSTM in Figure 4.48 draws on a broad set of momentum and technical indicators: RSI_14, MACD histogram, and volatility ratio rank among the most important features, reflecting a reliance on short-term technical signals. Return-based features and moving averages also contribute meaningfully, showing that the model balances price action with momentum cues. Bullish drivers are led by RSI_14 and recent returns; risk drivers flag MACD histogram and volatility as potential downside contributors. The BiLSTM in Figure 4.49 shows a far more concentrated structure: HL_range and Mom_10 account for almost all of the model's predictive influence, with most other variables contributing negligibly. This reflects a simplified decision process driven pri-

marily by price range expansion and medium-term momentum rather than the broader technical landscape. The contrast between the two figures mirrors what was seen across other coins: CNN-LSTM captures a wider set of market dynamics while BiLSTM focuses on a small number of dominant features.

Chapter 5

System Implementation

5.1 Development Environment

The development environment was chosen with performance, scalability, and reproducibility in mind. Given the computational demands of the hybrid CNN-LSTM + BiLSTM architecture, all model training was carried out on Kaggle Notebooks equipped with dual NVIDIA Tesla T4 GPUs. This setup provided enough memory and parallel processing capacity to train complex model combinations within a reasonable timeframe.

TensorFlow and Keras serve as the core deep learning frameworks. Their flexibility, strong community support, and native GPU compatibility made them the natural choice. The Keras Functional API in particular allowed the hybrid architecture to be built with multiple parallel branches operating simultaneously. For data processing and feature engineering, Pandas, NumPy, and Scikit-learn handled the heavy lifting: large-scale time-series operations, preprocessing pipelines, normalization, dataset splitting, and evaluation utilities.

The backend was built with FastAPI, chosen for its speed, asynchronous request handling, and straightforward support for building RESTful APIs. This made it a practical fit for a system where real-time prediction requests need to be handled with low latency and clean JSON responses.

The frontend was developed using React combined with Vite for fast builds and hot module replacement during development. Tailwind CSS was used for styling, enabling a clean, modern interface without a large custom CSS footprint. Recharts was integrated for rendering prediction charts and performance visualizations. Version control ran through Git, and Visual Studio Code served as the primary development environment. [15, 16, 17]

Table 5.1: Technology Stack of the Proposed System

Component	Technology / Version
Training Platform	Kaggle Notebooks — GPU T4 x2 (15 GB VRAM)
Deep Learning Framework	TensorFlow 2.x / Keras
Data Processing	Pandas, NumPy, Scikit-learn
ARIMA Forecasting	pmdarima (auto.arima)
Backend Framework	FastAPI + Uvicorn
Frontend Framework	React 18 + Vite
Styling	Tailwind CSS 3
Animation	Framer Motion 11
Charts	Recharts 2
HTTP Client	Axios
Version Control	Git

5.2 Data Collection and Feature Engineering

Data collection and feature engineering are what make or break a prediction system like this. A model is only as good as the data it learns from, and in a domain as noisy and volatile as cryptocurrency, how you prepare that data matters enormously. Historical price data is collected using the Yahoo Finance API through the `yfinance` library, pulling OHLCV values: Open, High, Low, Close, and Volume. These are the foundational inputs for any financial time-series model. The data fetching module includes retry mechanisms and fallback handling so that temporary API failures do not break the pipeline.

Alongside market data, the Fear and Greed Index API supplies a numerical sentiment signal ranging from extreme fear to extreme greed. Because cryptocurrency prices are so strongly influenced by crowd psychology, this signal adds meaningful context that raw price data alone cannot capture. [22]

Feature Engineering Pipeline

Feature engineering is performed in `preprocessing.py` using pandas rolling operations. Thirty-one features are generated in total, covering returns, volatility, momentum, trend indicators, and sentiment. These features are designed to give the model visibility into both short-term price fluctuations and longer-term structural trends. Moving averages like SMA and EMA smooth out noise; RSI and MACD capture momentum and potential trend reversals; Bollinger Bands track volatility levels; and volume-based features reflect how actively the market is participating at any given time.

The dataset is then split chronologically to preserve temporal integrity and prevent data leakage. A `RobustScaler` handles normalization, which is particularly well-suited to

crypto data given its tendency toward extreme outliers. Finally, sliding window sequences of 30 days are generated to convert the time-series into a supervised learning format, where each sequence feeds the model a window of recent history from which it predicts what comes next.

The complete feature set is organized as follows:

Table 5.2: Feature Engineering Categories and Extracted Features for Cryptocurrency Prediction System

Category	Features	Description
Returns	LogReturn, Return_1d/3d/7d/14d	Multi-horizon returns
Volatility	Vol_10, Vol_30, HL_range	Rolling volatility
Moving Averages	SMA10/30/50/200	Trend smoothing indicators
Regime	Bull_regime, Trend_strength, MA_cross	Market regime detection
Momentum	RSI_6, RSI_14, Mom_5, Mom_10	Momentum indicators
Bollinger Bands	BB_position, BB_width	Volatility bands
MACD	MACD, Signal, Histogram	Trend reversal signals
Volume	OBV, Vol_ratio, Vol_change	Market participation
Price Position	Price_pos	Relative price location
Sentiment	FearGreed, FG_change	Market sentiment

5.3 Model Implementation

The hybrid model is implemented in *lstm_model.py* [15, 17, 21] using the Keras Functional API, which enables a parallel architecture with two main branches:

- **CNN-LSTM Branch (local + sequential pattern learning):** This branch handles both short-term and longer-term pattern learning. CNN layers act as feature extractors, identifying local trends, sudden spikes, and micro-patterns in the time-series data that would be difficult for a recurrent model to pick up directly.

The extracted features are then passed to LSTM layers, which model sequential dependencies and temporal relationships over time. This combination gives the branch a strong ability to understand how past price conditions evolve into future trends.

- **BiLSTM Branch (global dependency learning):** The BiLSTM branch introduces multi-head attention, allowing the model to analyze all time steps simultaneously rather than processing them sequentially. This makes it particularly effective at identifying long-range dependencies, including relationships between events that are far apart in time. In volatile cryptocurrency markets, where the effects of market events can show up with significant delays, this global perspective adds real predictive value.
- **Fusion Layer (feature integration):** Outputs from the CNN-LSTM and BiLSTM branches are concatenated in a fusion layer and passed through dense layers to produce a unified feature representation. This merging step is where the complementary strengths of both branches come together: local sequential learning from CNN-LSTM and global attentive learning from BiLSTM are combined into a single vector that captures a more complete picture of market dynamics.
- **Dual Output Heads (price regression + direction classification):** The model ends with two output heads. The first performs regression to predict actual price values. The second performs classification to predict the direction of price movement: STRONG UP, NEUTRAL, or STRONG DOWN. This dual-output design makes the system more actionable in practice, providing both a numerical forecast and a direction signal that users can act on directly.

Key Implementation Design

Each cryptocurrency is trained independently, preventing cross-contamination of patterns between assets. This coin-specific approach improves generalization by allowing the model to learn the unique characteristics of each asset, rather than averaging across them.

The label generation strategy uses dynamic quantile thresholds derived from training data:

- **STRONG_UP (Top 15% of 5-day forward returns):** Flags significant upward price movements and helps the model focus on genuine bullish trends rather than noise.
- **STRONG_DOWN (Bottom 15%):** Flags major downward movements, enabling the model to detect bearish conditions with equal sensitivity.

- **Neutral (Remaining data):** Covers stable or minor movements, ensuring balanced representation across all three directional classes.

Key Design Choices

- **RobustScaler over MinMaxScaler:** Cryptocurrency data is notorious for extreme outliers. RobustScaler, which is based on median and interquartile range rather than min and max values, is far less sensitive to these spikes. Events like the COVID crash or the FTX collapse can distort a MinMaxScaler’s output significantly; RobustScaler handles them without scaling the entire dataset around a single extreme value.
- **Focal Loss ($\gamma = 2.0$):** Class imbalance is a natural problem here: neutral market movements far outnumber strong directional moves in the training data. Focal Loss addresses this by down-weighting easy examples and concentrating learning effort on the harder, less common cases, which improves performance on STRONG_UP and STRONG_DOWN predictions specifically.
- **Confidence Threshold = 0.45:** Only predictions above this confidence threshold are treated as strong signals. This filters out low-certainty outputs and reduces noise in the final prediction stream.
- **Adaptive Thresholding (min 60th percentile confidence, 0.50):** This dynamic mechanism adjusts thresholds based on the distribution of prediction confidence scores. It ensures that only statistically meaningful predictions make it through, improving overall robustness.

5.4 Training on Kaggle

Training is handled through *train_all_coins.py*, which runs the complete training pipeline for all supported cryptocurrencies in a single automated process [18]. The pipeline is built to be reproducible and hands-off: once started, it trains both the hybrid model and the BiLSTM baseline for each coin, saves all outputs, and generates evaluation artifacts without requiring manual intervention between runs. Running on Kaggle’s GPU environment makes it possible to complete this process in a reasonable timeframe despite the complexity of the architecture.

Training Process Steps

1. **Deterministic seed initialization (SEED=42):** The first step is locking down randomness. A fixed seed is applied across NumPy, TensorFlow, and Python before

any training begins. This ensures that the same pipeline run on the same data will produce the same results, which is important both for debugging and for comparing experiments fairly.

2. **Per-coin independent training loops:** Each cryptocurrency is trained in its own loop, completely separate from the others. This prevents patterns from one asset from contaminating the model trained on another, and allows each model to specialize in the unique volatility and behavioral characteristics of its asset.
3. **Model evaluation after each training cycle:** Once a model finishes training, it is evaluated immediately using RMSE, MAE, MAPE, and directional accuracy metrics. This step confirms that the model meets a reasonable performance bar before it is saved and handed off to the deployment pipeline.
4. **Metric logging (JSON format):** All evaluation results are saved to structured JSON files. This makes it easy to track performance across coins and across experiments, and the files are directly usable during deployment and visualization without any additional transformation.
5. **Visualization generation (confusion matrices, DA charts):** The pipeline automatically generates confusion matrices and directional accuracy charts for each trained model. These visuals make it easier to diagnose where the model is performing well and where it struggles, particularly in distinguishing between directional classes.
6. **Export of all models and artifacts as ZIP files:** After training completes, all models, scalers, metrics, and visual outputs are packaged into a ZIP archive. This ensures the trained artifacts are portable and ready for deployment into the FastAPI backend without any additional steps.

Key Training Features

- **No data leakage (strict chronological split):** Training, validation, and test sets are divided in chronological order with no overlap. Future data never enters the training phase, which preserves the integrity of the time-series evaluation and produces honest performance estimates.
- **GPU acceleration (T4 x2):** Dual T4 GPUs on Kaggle substantially reduce training time, making full-scale experimentation across seven coins practical within a single session.

- **Independent coin-wise training:** Each asset gets its own dedicated training run, allowing the model to capture asset-specific behavior rather than averaging across assets in a way that might weaken predictions for any individual coin.
- **Automated comparison with BiLSTM baseline:** Both the hybrid model and a BiLSTM baseline are trained for every coin. This makes it straightforward to measure what the hybrid architecture actually adds over a simpler sequential model, with consistent metrics across all seven assets.

5.5 Backend Implementation (FastAPI)

The backend is implemented in `main.py` using FastAPI and follows a modular, service-based architecture [13]. It sits at the center of the system, handling all model inference, data processing, and communication between the frontend and the machine learning components. FastAPI's asynchronous request handling and high throughput make it well-suited for a real-time prediction system where multiple users might be querying the backend simultaneously.

Core Backend Design Features

- **Model Caching (Models loaded once and reused in memory):** Trained models are loaded into memory once at server startup and reused for every subsequent request. This avoids the overhead of reading from disk on each inference call, which keeps response times short and consistent.
- **Data Caching (300 seconds): Reduces redundant preprocessing overhead:** Recently processed data is cached for 300 seconds. If a new request arrives for the same asset within that window, the preprocessed data is served from cache rather than fetched and reprocessed from scratch. This significantly reduces API load during active use.
- **Graceful Degradation (If model unavailable, Kaggle offline results are returned):** If a trained model fails to load or becomes unavailable for any reason, the backend falls back to precomputed offline results. This keeps the system operational during demos and avoids complete failure when something goes wrong at runtime.
- **ARIMA Blending (Final forecast = 0.6 (LSTM) + 0.4 (ARIMA)):** The final prediction output blends deep learning forecasts with a traditional ARIMA component in a 60/40 ratio. This hybrid blending improves stability, particularly

during periods of high volatility where purely neural outputs can fluctuate more than is useful for decision-making.

- **Custom Loss Handling (Focal loss registered via custom_objects in TensorFlow):** Because the model uses focal loss for directional classification, this custom loss function must be registered when loading the model from disk. This ensures the model restores correctly and behaves consistently with how it was trained.

API Behavior

The backend handles three core categories of request:

- **Real-time predictions:** Incoming data is preprocessed and routed through the trained hybrid model, returning price forecasts and directional signals with minimal delay.
- **Historical data requests:** Past market data and prediction records are retrieved and formatted for visualization in the frontend dashboard.
- **Model evaluation responses:** Performance metrics and evaluation results are returned in a structured format, allowing users to assess how the model is performing on a given asset at any point.

5.6 Frontend Implementation (React)

The frontend is built with React 18 and Vite, following a component-based architecture [14]. It serves as the user-facing layer of the system, designed to surface predictions and market analytics in a way that is clear and accessible without requiring the user to understand anything about the underlying models. React's modular component structure keeps the codebase maintainable, and Vite speeds up development through fast builds and hot module replacement.

Architecture Design

Reusable UI components: The interface is built from a set of modular components that can be composed across different pages:

- GlassCard
- PredictionCard
- MetricsPanel

- SentimentPanel

Pages:

- Landing Page
- Dashboard
- Model Information Page

API Handling

Backend communication is managed through the useApi.js hook using Axios:

- On startup, the hook runs a backend health check to confirm the server is available before making any prediction requests.
- If the backend is unreachable, the system automatically falls back to precomputed Kaggle results, keeping the application functional even without a live backend connection.
- This fallback mechanism ensures the system remains fully demonstrable in offline or demo environments.

Visualization System

The ForecastChart component displays three overlaid data series:

- **Actual Price (white line):** Real market data, providing the ground truth against which users can compare model predictions.
- **CNN-LSTM Prediction (green dashed):** The hybrid model's forecast, showing expected future price direction.
- **BiLSTM Prediction (amber dashed):** The baseline model's forecast, displayed for direct comparison.

Standard line charts are used in place of area charts to keep overlapping prediction lines legible. When multiple forecast series are displayed together, area fills tend to obscure each other; clean lines make it easier for users to track each model's output independently.

Chapter 6

Testing & Evaluation

6.1 Evaluation Metrics

Table 6.1: Evaluation Metrics Used for Model Performance Assessment

Metric	Formula	Description
MAPE	$\text{mean}(\text{actual} - \text{pred} / \text{actual})$	Percentage error
RMSE	$\sqrt{\text{mean}(\text{error}^2)}$	Prediction deviation
DA (Strong)	Accuracy on extreme moves	Directional accuracy
DA (Conf+Strong)	Accuracy on confident predictions	Main metric
F1 Macro	Mean F1 score	Balanced classification
Confusion Matrix	TP, FP, TN, FN	Classification breakdown

6.2 Results

The results demonstrate that the proposed hybrid model holds up well across multiple cryptocurrencies in both regression and classification tasks. The model captures complex patterns in highly volatile financial data effectively enough to maintain low prediction error on MAPE while achieving meaningful directional accuracy, particularly for Bitcoin where results are strongest. This combination, low numerical error alongside reliable directional signals, is what makes the system practically useful rather than just theoretically interesting.

Performance does vary across assets, and that variation is expected. Differences in volatility, liquidity, and how much historical data is available all affect how much the

model can learn. But the overall consistency across seven quite different assets is a reasonable indicator that the hybrid architecture generalizes rather than overfitting to any one coin’s behavior.

Table 6.2: Performance Results of the Proposed Model Across Different Cryptocurrencies

Coin	DA Strong	DA Conf	MAPE	Accuracy	F1 Macro
BTC	61.7%	70.8%	1.68%	98.32%	0.5832
ETH	67.0%	50.0%	2.79%	97.21%	0.4011
BNB	63.2%	56.5%	1.97%	98.03%	0.3871
XRP	50.9%	47.6%	2.92%	97.08%	0.4913
ADA	61.7%	60.4%	3.48%	96.52%	0.5978
SOL	72.1%	58.8%	3.10%	96.90%	0.4189
DOGE	57.4%	48.4%	3.60%	96.40%	0.5513

6.3 Comparison with BiLSTM

Comparing the hybrid model against the BiLSTM baseline makes the value of the hybrid approach concrete. The baseline is a reasonable model in its own right: it captures sequential dependencies well and performs competitively on regression metrics. But it lacks the ability to simultaneously extract local patterns through convolution and attend globally through BiLSTM, which limits its ceiling.

For Bitcoin, Binance Coin, and Cardano, the hybrid CNN-LSTM + BiLSTM model outperforms the baseline clearly, particularly on directional accuracy metrics. On some altcoins, the BiLSTM baseline actually comes out slightly ahead, which is a useful reminder that simpler models can generalize better when data is noisy or limited. It reinforces the point that model selection should be informed by the specific asset and its characteristics rather than defaulting to the most complex available option.

6.4 Discussion

6.4.1 BTC Performance

Bitcoin achieves 70.8% DA (Conf + Strong), the strongest result across all seven coins. Several factors work together to produce this:

- **High liquidity:** Bitcoin has the largest trading volume of any cryptocurrency, which translates into more stable and predictable price dynamics and less susceptibility to sudden manipulation or thin-order-book moves.

- **Stable long-term patterns:** Compared to most altcoins, BTC’s price history shows more consistent structural trends, giving the model more reliable patterns to learn from.
- **Institutional trading behavior:** The presence of institutional investors introduces a degree of structured market behavior that reduces the impact of purely speculative noise.
- **Large training dataset (8+ years):** A longer historical record gives the model substantially more data to train on, which helps it generalize to new market conditions rather than overfitting to a short historical window.

Table 6.3: Comparative Analysis of CNN-LSTM + BiLSTM Model with BiLSTM Baseline

Coin	CNN DA Conf	BiLSTM DA Conf	CNN MAPE	BiL MAPE	Winner
BTC	70.8%	37.5%	1.68%	1.67%	CNN
ETH	50.0%	54.8%	2.79%	2.82%	BiLSTM
BNB	56.5%	43.5%	1.97%	1.98%	CNN
XRP	47.6%	57.1%	2.92%	2.89%	BiLSTM
ADA	60.4%	56.2%	3.48%	3.49%	CNN
SOL	58.8%	70.0%	3.10%	3.09%	BiLSTM
DOGE	48.4%	51.6%	3.60%	3.55%	BiLSTM

6.4.2 Altcoin Performance

Lower performance on altcoins is a predictable outcome given the structural differences between those assets and Bitcoin:

- **High volatility and speculative behavior:** Altcoin prices can move sharply in response to social media activity, exchange listings, or even individual influencer posts, introducing noise that is very hard to model.
- **External event dependency:** Price moves for assets like XRP and ADA are frequently driven by regulatory news, project partnerships, or network upgrades rather than market-wide trends.
- **BTC-centric sentiment bias:** The Fear and Greed Index is designed primarily around Bitcoin market behavior, which reduces its effectiveness as a sentiment signal for altcoins with different driver dynamics.

- **Shorter historical data (SOL, DOGE):** Assets with limited historical records give the model less to learn from, which makes generalization harder and results more variable.

6.4.3 Feature Importance Insight

One of the most practically useful observations from the experimentation phase was how much feature selection matters. Early in the project, several sentiment-related features were included alongside the Fear and Greed Index: Google Trends data and Bitcoin dominance metrics, among others. These features appeared informative in isolation, but in practice they introduced redundancy. They were capturing overlapping market signals, and the model struggled to separate meaningful patterns from the noise created by correlated inputs.

After removing these redundant features, Bitcoin prediction accuracy improved from around 50% to 70.8%. That is a substantial jump, and it did not come from a model architecture change or more training data. It came from cleaning up the input feature set.

This finding makes an important broader point: more features is not the same as better features. In financial deep learning, and especially in the noisy, high-variance environment of cryptocurrency markets, the quality and independence of input signals matter far more than the quantity. Careful feature selection is not just a preprocessing step; it is one of the most impactful design decisions in the entire pipeline.

Chapter 7

Conclusion & Future Work

This project delivers a complete, end-to-end hybrid CNN-LSTM + BiLSTM system for cryptocurrency price prediction. From data collection through feature engineering, model training, backend deployment, and interactive frontend visualization, the system covers the full pipeline and puts it into practice across seven real cryptocurrency assets.

The architectural core of the work is the combination of three complementary learning mechanisms. CNN layers extract short-term patterns and local trends from price data. LSTM layers model how those patterns evolve sequentially over time. BiLSTM with attention captures global relationships across the entire input sequence, identifying dependencies that a purely sequential model would miss. Together, these components give the system a much richer view of market dynamics than any single model could achieve.

The practical side of the project is equally important. The FastAPI backend and React frontend turn a research model into a usable tool: one that generates real-time predictions, visualizes forecasts interactively, and exposes evaluation metrics that help users understand how much to trust any given prediction. A system that works in theory but fails in deployment is not a complete solution, and this project treats deployment as a first-class concern rather than an afterthought.

The results confirm that hybrid deep learning architectures meaningfully outperform traditional single-model approaches in financial forecasting, particularly on directional accuracy and cross-asset robustness. The performance gap is clearest on Bitcoin, where the hybrid model achieves 70.8% confident directional accuracy, but the system also demonstrates reasonable generalization across a diverse set of assets.

7.1 Key Achievements

The results validate that hybrid architecture significantly outperforms traditional sequential models in financial forecasting tasks.

- 70.8% directional accuracy on BTC, the strongest result achieved in this experimental setup.
- A 33.3% improvement over the BiLSTM baseline on Bitcoin’s confident directional accuracy metric.
- Full-stack deployment combining a React frontend, FastAPI backend, and complete ML pipeline.
- Demonstrated generalization across seven distinct cryptocurrency assets.

7.2 Limitations

- The system currently supports only daily timeframe predictions. Intraday forecasting, which would be needed for high-frequency trading applications, is not supported.
- The sentiment model is primarily calibrated around Bitcoin market behavior, which reduces its effectiveness as a signal for altcoins that respond to different drivers.
- Performance on altcoins is inconsistent, reflecting the inherently harder prediction problem posed by assets with higher volatility, less historical data, and stronger sensitivity to external events.
- There is no live trade execution module. The system is designed as a decision-support tool rather than an automated trading agent.
- Confidence calibration could be improved; the current confidence scores are informative but would benefit from more rigorous probabilistic calibration.

7.3 Future Work

7.3.1 Data Enhancements

- Integrating real-time Twitter and Reddit sentiment analysis would provide a richer, faster-moving signal for short-term prediction, especially for altcoins whose prices are more responsive to social media dynamics.
- Adding on-chain blockchain metrics such as total value locked, hash rate, and transaction volume would bring fundamental network health indicators into the feature set alongside the existing technical and sentiment features.

7.3.2 Model Improvements

- A reinforcement learning-based trading agent could be built on top of the existing prediction outputs to support automated decision-making with explicit reward signals.
- Attention visualization tools would improve interpretability, making it easier to show users which time steps and features are driving each prediction.
- MLflow experiment tracking would support more rigorous model versioning and reproducibility across training runs.

7.3.3 System Enhancements

- Deploying to cloud infrastructure on AWS or GCP would improve scalability, up-time, and the ability to serve multiple users concurrently.
- A real-time streaming data pipeline would allow predictions to update continuously as new price data arrives rather than on a request-response basis.
- A mobile application version would make the system accessible to a broader range of users, including those who primarily operate from phones rather than desktop environments.

7.3.4 Advanced Extensions

- Support for intraday timeframes such as 1-hour or 4-hour candles would open the system to day trading use cases, which currently fall outside its scope.
- Integration with trading APIs would allow the system to move from a decision-support role into automated trade execution for users who want that capability.
- A portfolio optimization module would allow users to manage risk across multiple positions rather than evaluating individual assets in isolation.

7.4 Closing Statement

This Final Year Project presents a comprehensive, practically deployed AI-driven financial prediction system that brings together advanced deep learning methodology and real-world engineering in a single working platform. By integrating data collection, feature engineering, hybrid model design, backend services, and frontend visualization into one coherent system, the project demonstrates what it actually takes to move a machine learning idea from concept to deployment.

The hybrid CNN-LSTM + BiLSTM architecture at the center of this work is well-suited to the challenge it was designed for. Cryptocurrency markets are complex, volatile, and driven by dynamics that no single model architecture can fully capture. By distributing learning responsibilities across convolutional, recurrent, and attention-based components, the system approaches the problem from multiple angles simultaneously. The results, particularly the 70.8% confident directional accuracy on Bitcoin, show that this multi-perspective approach pays off in measurable ways.

The integration of sentiment analysis and a complete deployment stack adds practical value that a standalone model would lack. Predictions are only useful if users can access them easily and interpret them correctly, and the system is designed with that reality in mind.

There are real limitations, and this report has tried to be honest about them. Altcoin performance is inconsistent, intraday prediction is not supported, and confidence calibration could be sharper. These are not reasons to dismiss the work; they are the starting points for the next round of improvements.

As a foundation, this system is extensible in the directions that matter: richer data sources, stronger models, cloud-scale deployment, and eventually automated trading integration. The work points toward a future where intelligent, explainable financial forecasting tools are not just available to institutional players but accessible to individual researchers and traders who want to make more informed decisions.

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